Policy Optimization: From CPI to PPO

Conservative Updates, Trust Regions, Advantage Estimation & Clipping

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Part I — Main Sources

The Policy Gradient Theorem

Sutton et al. (1999)

The foundation that proved you can optimize a parameterized policy directly by estimating the gradient of the expected return.

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\right]$$



Neuro-Dynamic Programming

Bertsekas & Tsitsiklis (1996), Ch. 6

Formalized Approximate Policy Iteration (API). Chapter 6 is historically significant here because it exposed the fundamental instability of using function approximators (like neural networks) in reinforcement learning.

Conservative Policy Iteration

Kakade & Langford (2002)

Established the theoretical foundation for monotonic policy improvement using the performance difference lemma and mixture policies.

Trust Region Policy Optimization

Schulman et al. (2015)

Scaled CPI's concepts to deep neural networks by using a KL-divergence constraint to enforce a "trust region" during policy updates, preventing destructively large steps.

$$J(\pi_{new}) \geq L_{\pi_{old}}(\pi_{new}) - \frac{4\gamma\epsilon}{(1-\gamma)^2}\max_s D_{KL}(\pi_{old}(\cdot|s)\|\pi_{new}(\cdot|s))$$

Approximately Optimal Approximate Reinforcement Learning

Kakade & Langford
ICML 2002

Introduces **Conservative Policy Iteration (CPI)**

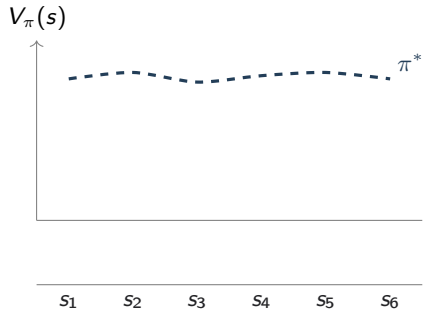
Setting & Assumptions

A finite MDP $(S, D, A, \mathcal{R}, P)$:

- **Finite** state and action spaces: $|S|, |A| < \infty$.
- **Bounded rewards**: $\mathcal{R} : S \times A \to [0, R]$.
- **Discounted**: $\gamma \in [0, 1)$.
- **Unknown dynamics**: $P(s'; s, a)$ is not given — we can only sample transitions (model-free).
- **Start-state distribution** D: the goal is

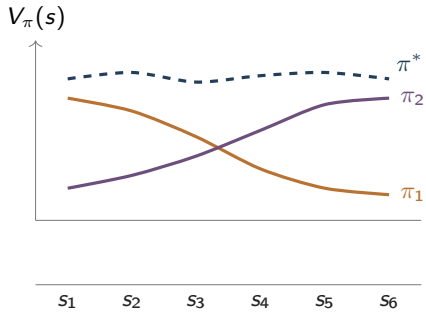
$$\eta_D(\pi) = \mathbb{E}_{s_0 \sim D}\left[V_\pi(s_0)\right]$$

The Start-State Distribution D



Exact: π^* on top everywhere
$\Rightarrow$ D irrelevant.

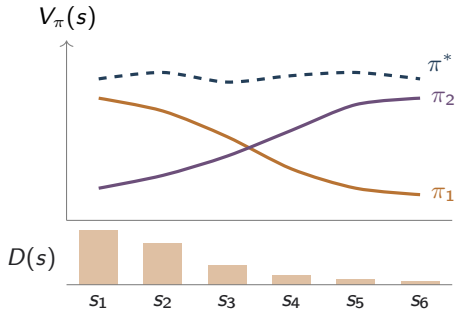
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Approximate: curves cross
restricted class $\{\pi_\theta\}$ · limited data

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Goal (Sec. 2): rank by D

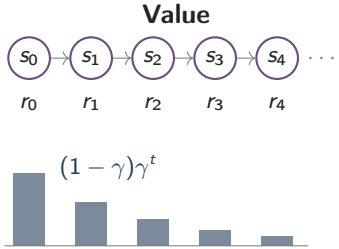
$$\eta_D(\pi) = \sum_s D(s)\, V_\pi(s)$$

Here π_1 wins.

D **comes from the application:** robot start poses, game openings, user prompts.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$



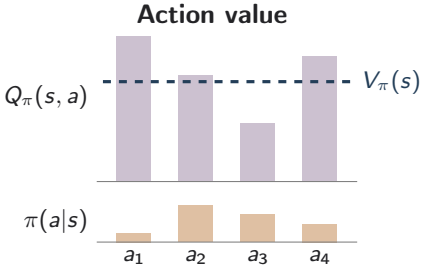
Weights sum to 1, so V is an average of rewards: $V \in [0, R]$.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$

$$Q_\pi(s,a) = (1-\gamma)\mathcal{R}(s,a) + \gamma\,\mathbb{E}_{s'}\big[V_\pi(s')\big]$$

$$V_\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)}\big[Q_\pi(s,a)\big]$$



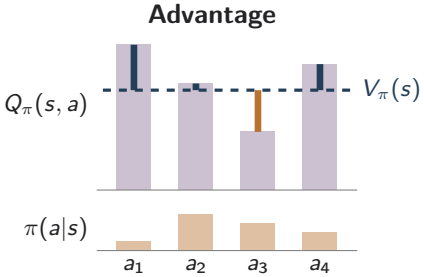
Take a first, then follow π.
V_π = bars averaged with weights $\pi(a|s)$.

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$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$
$$\mathbb{E}_{a\sim\pi}\big[A_\pi(s,a)\big] = 0, \quad \mathbb{E}_{a\sim\pi'}\big[A_\pi(s,a)\big] \neq 0$$



A = gap to the line. Weighted by π the gaps cancel.
A π' that favors the **positive** gaps gets $\mathbb{E}_{\pi'}[A_\pi] > 0$.

Notation

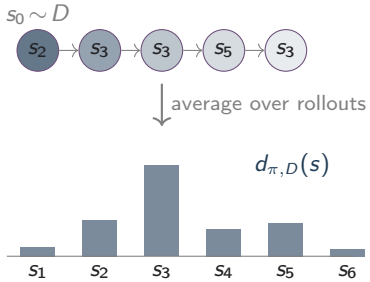
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$$d_{\pi,D}(s) = (1-\gamma)\sum_t \gamma^t \Pr(s_t = s;\, \pi, D)$$

Discounted visitation



Where π spends its time from $s_0 \sim D$, early steps weighted more.
Not the same as D (only s_0).

Notation

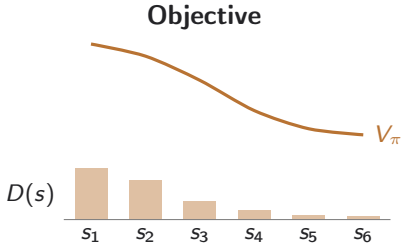
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$$\eta_D(\pi) = \mathbb{E}_{s\sim D}\big[V_\pi(s)\big]$$



Average V_π over start states only.
One number to rank policies.

Three Questions Any Approximate Method Should Answer

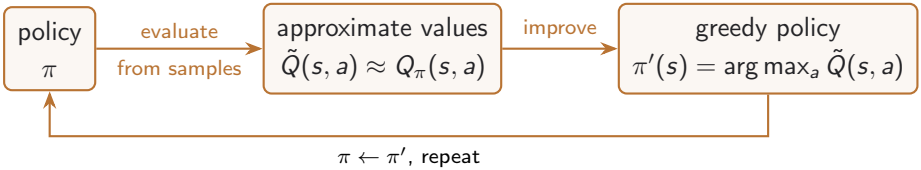
1. Is there a **performance measure** that is guaranteed to **improve at every step**?
2. How hard is it to **verify** that a given update improves this measure?
3. What **performance level** is reached after a **reasonable number** of updates?

Note: not "what is the asymptotic policy?" — only policies reachable in reasonable time.

Approximate value functions
Weak answers to all three.

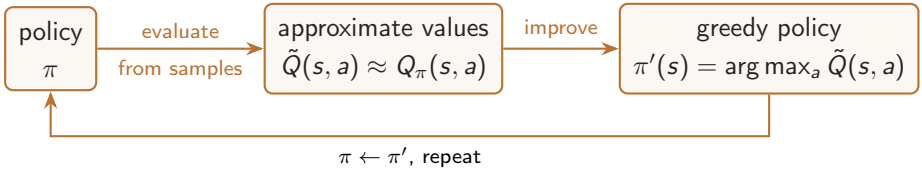
Policy gradients
Good answer to (1), poor answers to (2) and (3).

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Example: Q-learning with a function approximator $\tilde{Q}_w$

$$w \leftarrow w + \alpha \left(y_t - \tilde{Q}_w(s_t, a_t)\right) \nabla_w \tilde{Q}_w(s_t, a_t)$$

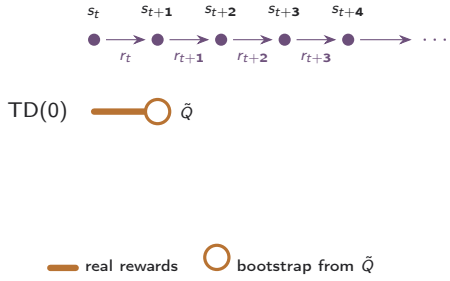
$$y_t = (1-\gamma)\, r_t + \gamma \max_{a'} \tilde{Q}_w(s_{t+1}, a')$$

The max is the greedy step, so evaluate and improve happen together.
The target y_t is one choice of many $\rightarrow$ next slide.

How Is $\tilde{Q}$ Estimated? From TD(0) to Monte Carlo

TD(0): one reward, then bootstrap

$$y_t^{(1)} = (1-\gamma)\, r_t + \gamma\, \tilde{Q}(s_{t+1}, a_{t+1})$$



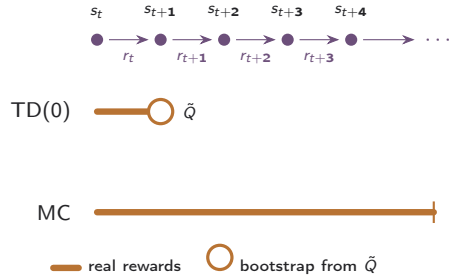
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Monte Carlo: the whole observed return

$$y_t^{(\infty)} = (1-\gamma) \sum_{k \geq 0} \gamma^k r_{t+k}$$



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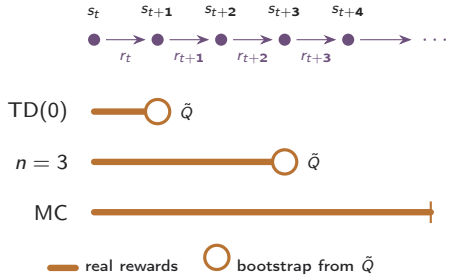
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n-step: in between

$$y_t^{(n)} = (1-\gamma) \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n\, \tilde{Q}(s_{t+n}, a_{t+n})$$



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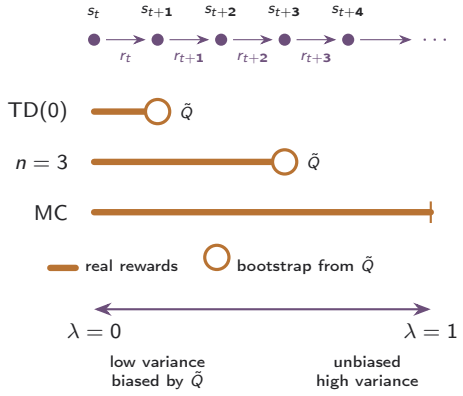
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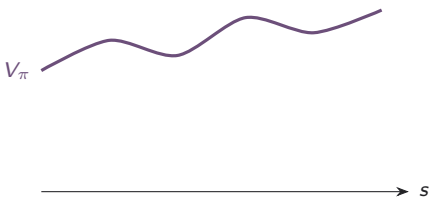
$$y_t^{(n)} = (1-\gamma) \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n\, \tilde{Q}(s_{t+n}, a_{t+n})$$

TD(λ): average over all n

$$y_t^{\lambda} = (1-\lambda) \sum_{n \geq 1} \lambda^{n-1}\, y_t^{(n)}$$



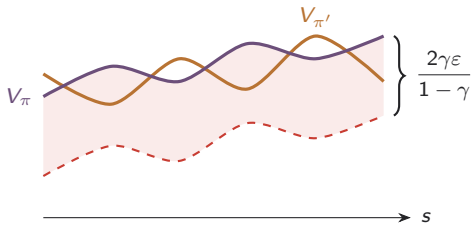
Gap 1: Approximate Value Functions Give No Improvement Guarantee



Approximation error, measured **everywhere** (ℓ_∞):

$$\varepsilon = \max_s \left| \tilde{V}(s) - V_\pi(s) \right|$$

Gap 1: Approximate Value Functions Give No Improvement Guarantee



$V_{\pi'}$ may sit anywhere above the dashed line — **also below** V_π.

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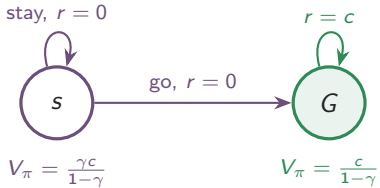
With π' greedy w.r.t. $\tilde{V}$ (Bertsekas & Tsitsiklis):

$$V_{\pi'}(s) \;\geq\; V_\pi(s) - \frac{2\gamma\varepsilon}{1-\gamma} \qquad \forall s$$

Only "won't get **much worse**" — never "gets better".

Gap 1: The Bound Is Tight

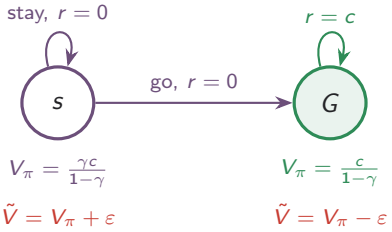
1. π goes to G and collects c forever.



π = go. Set $c = 2\varepsilon$.

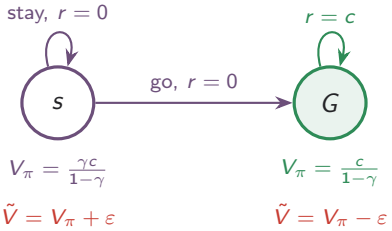
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1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.



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Gap 1: The Bound Is Tight



π = go. Set $c = 2\varepsilon$.

1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.
3. Greedy w.r.t. $\tilde{V}$ at s:

$$\underbrace{\gamma\big(V_\pi(s) + \varepsilon\big)}_{\text{stay}} = \underbrace{\gamma\big(V_\pi(G) - \varepsilon\big)}_{\text{go}}$$

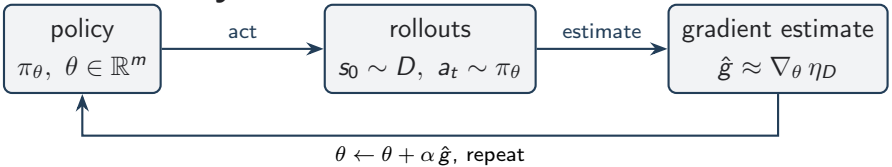
a tie, since $V_\pi(G) - V_\pi(s) = c = 2\varepsilon$, so π' may stay forever: $V_{\pi'}(s) = 0$.

At s: $V_\pi(s) - V_{\pi'}(s) = \dfrac{\gamma c}{1-\gamma} = \dfrac{2\gamma\varepsilon}{1-\gamma}$ — exactly the bound.

At G: one action only, so $\pi' = \pi$ and the loss is 0.

The bound is a max over states; s attains it.

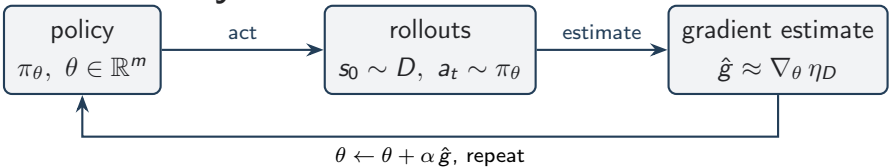
Reminder: Policy Gradient Methods



$$\nabla_\theta \eta_D = \frac{1}{1-\gamma} \sum_{s,a} d_{\pi,D}(s)\, \nabla_\theta \pi(a;s)\, Q_\pi(s,a)$$

Small steps uphill on η_D itself $\Rightarrow$ a good answer to question 1.

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Example: REINFORCE with a Gibbs (softmax) policy

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}, \qquad \hat{g} \ \propto \ \sum_t \nabla_\theta \log \pi_\theta(a_t;s_t)\, \hat{Q}_t$$

$\hat{Q}_t$: return observed after step t. Same policy class as Example 2.

▸ Gibbs policy

Policy Gradient Is Stochastic Gradient Ascent

	Stochastic gradient descent		Policy gradient
objective	$\min_w \; J(w) = \mathbb{E}_X\big[f(w, X)\big]$	$\leftrightarrow$	$\max_\theta \; \eta_D(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\big]$
looking for	$\nabla J(w) = \mathbb{E}\big[\nabla_w f(w, X)\big] = 0$	$\leftrightarrow$	$\nabla_\theta \, \eta_D(\theta) = 0$

Policy Gradient Is Stochastic Gradient Ascent

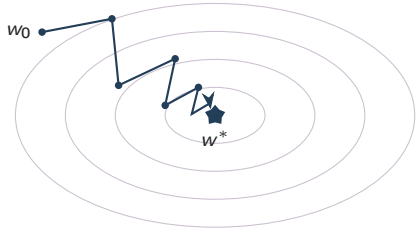
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looking for	$\nabla J(w) = \mathbb{E}\big[\nabla_w f(w, X)\big] = 0$	$\leftrightarrow$	$\nabla_\theta \, \eta_D(\theta) = 0$
one sample	$x_k \sim X$	$\leftrightarrow$	rollout τ_k: $s_0 \sim D, \;\; a_t \sim \pi_{\theta_k}$
update	$w_{k+1} = w_k - a_k \, \nabla_w f(w_k, x_k)$	$\leftrightarrow$	$\theta_{k+1} = \theta_k + a_k \, \hat{g}_k$

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update	$w_{k+1} = w_k - a_k \, \nabla_w f(w_k, x_k)$	$\leftrightarrow$	$\theta_{k+1} = \theta_k + a_k \, \hat{g}_k$
noise	$\eta_k = \nabla_w f(w_k, x_k) - \nabla J(w_k)$	$\leftrightarrow$	$\eta_k = \hat{g}_k - \nabla_\theta \, \eta_D(\theta_k)$

What we know about SGD carries over. Twist: samples come from π_θ itself, yet the log-derivative trick (REINFORCE) keeps $\hat{g}_k$ **unbiased**. ▸ Why?

SGD Convergence: the Convex, Smooth Case



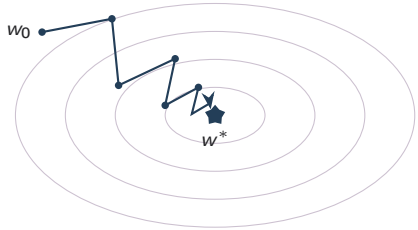
noisy steps, shrinking step size

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\rightarrow$ one minimizer, and bounds on the rate

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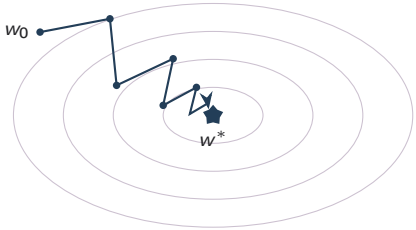
Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\rightarrow$ steps point the right way on average:
the estimate converges to w^*

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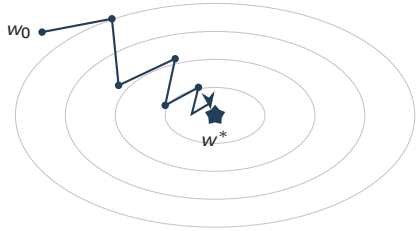
Finite variance and **step sizes**

$$\mathbb{E}[\eta_k^2 \mid \mathcal{H}_k] < \infty$$

$$\sum_k a_k = \infty, \quad \sum_k a_k^2 < \infty$$

$\rightarrow$ steps reach any distance, summed noise stays finite:
noise is filtered out relative to the signal

SGD Convergence: the Convex, Smooth Case



noisy steps, shrinking step size

$w_k \to w^*$ **almost surely**

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\to$ one minimizer, and bounds on the rate

Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\to$ steps point the right way on average:
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Sample Complexity: What Theory Guarantees

$$N(\epsilon, \delta) = \min \left\{ N \in \mathbb{N} \;\middle|\; \mathbb{P}\big(\|w_k - w^*\| \leq \epsilon\big) \geq 1 - \delta \;\text{ for all } k \geq N \right\}$$

samples needed to be ϵ-close with probability $\geq 1 - \delta$ (or in value: $\mathbb{P}(J(w_k) - J(w^*) \leq \epsilon) \geq 1 - \delta$)

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policy gradients with neural nets →

Convex, smooth

global optimum: $N(\epsilon, \delta) = O\big(\sigma^2/(c_1^2 \epsilon^2 \delta)\big)$

Non-convex, smooth

stationary point only: $\mathbb{E}\|\nabla J(w_R)\| \leq \epsilon$ after $K = O(\sigma^2/\epsilon^4)$, w_R: a random iterate, $R \leq K$

Non-convex, non-smooth

weaker still: near-stationary points, special classes

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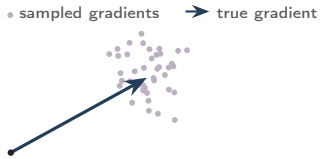
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Non-convex, non-smooth	weaker still: near-stationary points, special classes

Still useful: the convex case shows **what drives** N — the noise σ^2 and the curvature — and that carries over.

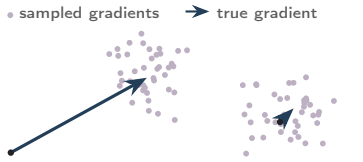
The Signal-to-Noise Reality



large $\|\nabla J\|$:
direction is clear

- Finite variance is enough for convergence *in the limit*.

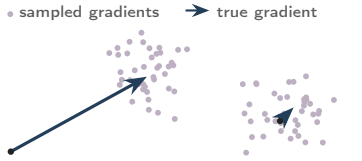
The Signal-to-Noise Reality



- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

$$n \gtrsim \sigma^2/\|\nabla J\|^2$$

The Signal-to-Noise Reality



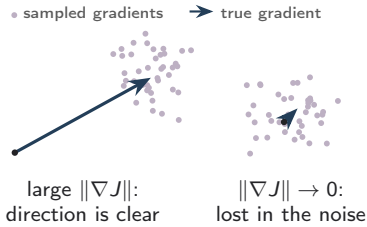
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- **Link to curvature**: strong convexity gives $\|\nabla J(w)\| \geq c_1\|w - w^*\|$, so at distance ϵ:

$$n \gtrsim \sigma^2/(c_1\epsilon)^2 \quad = \text{the convex-row } N \text{ (without } 1/\delta)$$

The Signal-to-Noise Reality



Next: two MDPs with an (almost) zero gradient.

- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

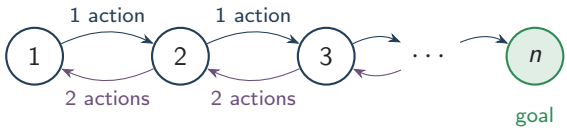
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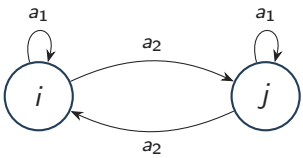
- Flat (c_1 small) or $\|\mathbb{E}[\nabla_w f(w, X)]\| \to 0$: noise dominates, **intractable** n.

Gap 2: Policy Gradients — Example 1 (Exploration)



- Uniform random policy: expected time to goal $= 3(2^n - n - 1) \approx 10^{15}$ for $n = 50$.
- Before reaching the goal, every sensible gradient estimate is 0: right *magnitude*, but **no direction**.
- Importance sampling doesn't help: weights would be exponentially large.

Gap 2: Policy Gradients — Example 2 (Plateaus)

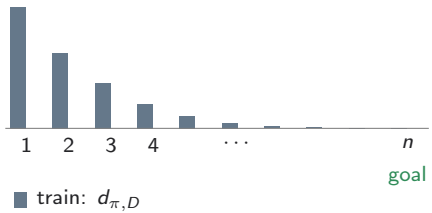


Gibbs policy: $\pi(a; s) \propto \exp(\theta_{sa})$

- Start: $\rho(i) = 0.8,\ \rho(j) = 0.2$.
- Raising the self-loop at i **starves** state j.
- $\rho(j)$ drops to $\approx 10^{-7} \Rightarrow$ nothing is learned at j.
- Result: an extremely **flat plateau** (on a 10^7 time scale).

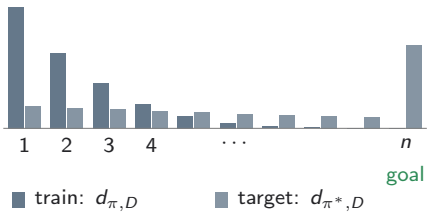
Root cause: η_D weighs improvements by $d_{\pi,D}$ — it is **insensitive to improvement at unlikely states**, even when those states are needed for near-optimal performance.

Gap 2 as Covariate Shift



Train: samples come from where the **current** π goes.

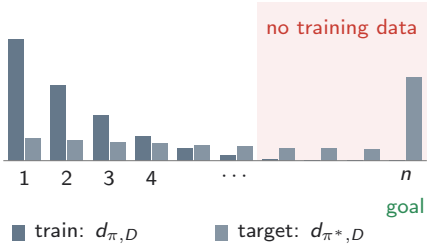
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Gap 2 as Covariate Shift

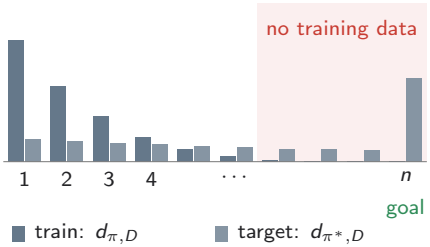


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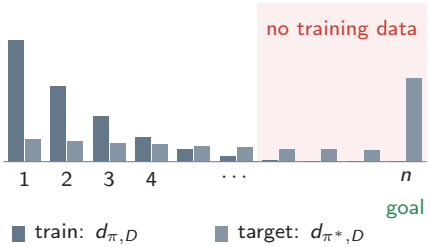
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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Gap 2 as Covariate Shift



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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Instead: change *where the data comes from*: train from μ, still judged by η_D.

$\rightarrow$ restart distribution μ (next slide)

The Two Fixes

1. Conservative update

Don't jump to the greedy policy — mix:

$$\pi_{\text{new}}(a; s) = (1 - \alpha)\,\pi(a; s) + \alpha\,\pi'(a; s)$$

$\alpha \in [0, 1]$ acts as a **step size**. $\alpha = 1$ is greedy policy iteration.

$\rightarrow$ addresses Gap 1 (greedy jumps)

2. Restart distribution μ

Optimize a measure that weighs states more uniformly:

$$\eta_\mu(\pi) = \mathbb{E}_{s\sim\mu}\big[V_\pi(s)\big]$$

μ is "exploratory"; the agent can restart from $s \sim \mu$ (weaker than a generative model*).

D: deployment (given) · μ: training resets (chosen)

$\rightarrow$ addresses Gap 2 (exploration)

- Any optimal policy maximizes both η_μ and η_D — but within a restricted class they can differ.

*Generative model (Kearns et al., 1999): sample $s' \sim P(\cdot \mid s, a)$ at *any* (s, a) you choose. A μ-restart only resets to a *random* $s \sim \mu$. Access, strongest to weakest: full P > generative model > μ-restart > one trajectory, no resets.

Roadmap: From the Two Fixes to the Three Answers

the two fixes

1. Conservative update
mix with a step α

2. Restart distribution
measure η_μ,
restarts $s \sim \mu$

results (next slides)

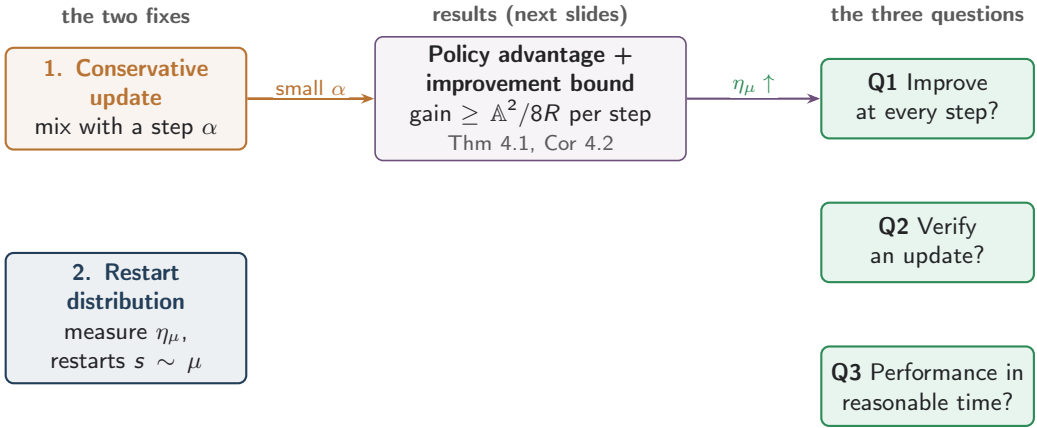
the three questions

Q1 Improve at every step?

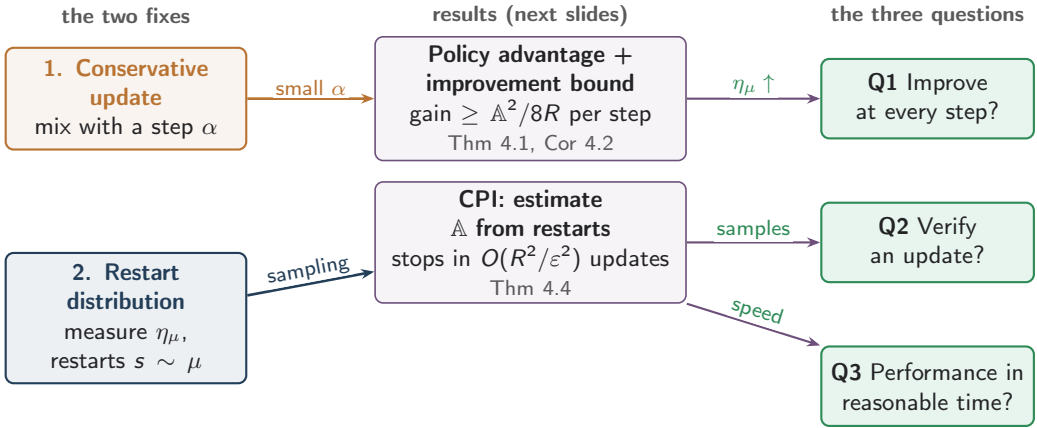
Q2 Verify an update?

Q3 Performance in reasonable time?

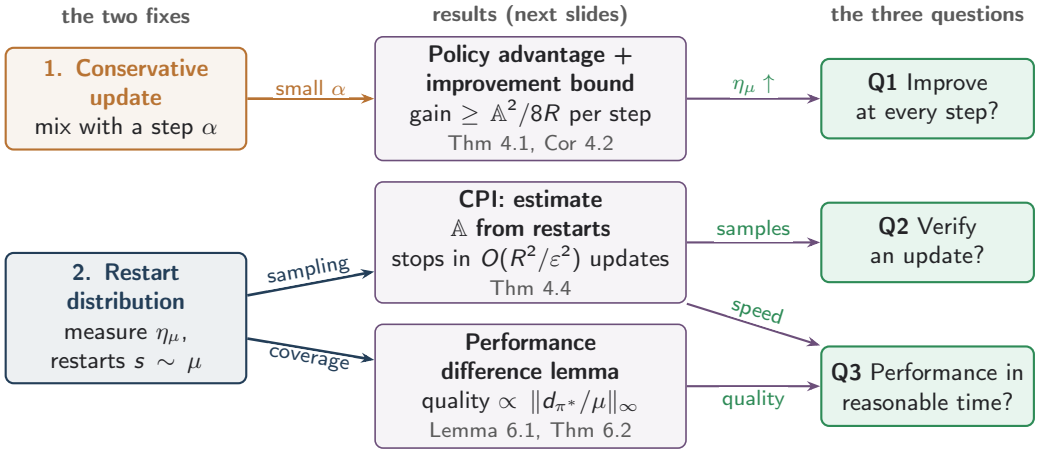
Roadmap: From the Two Fixes to the Three Answers



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Warm and cool arrows: where each fix enters. μ sets the **quality**, not the **speed**.

The Performance Difference Lemma

Per-state advantage of a policy π' over π — one step of π' at s, then back to π:

$$\bar{A}(s) \;=\; \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Lemma 6.1. For any π, π' and start distribution μ:

▸ Second equality

$$\eta_\mu(\pi') - \eta_\mu(\pi) \;=\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi',\mu}}\big[\bar{A}(s)\big] \;=\; \mathbb{E}\Big[\textstyle\sum_t \gamma^t\,\bar{A}(s_t)\;\Big|\; s_0\sim\mu,\ \pi'\Big]$$

The Performance Difference Lemma

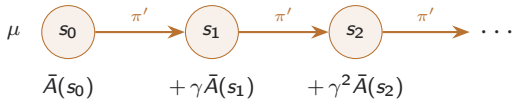
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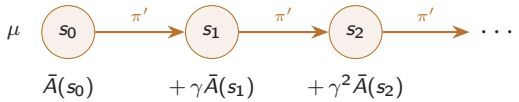
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- **Exact** — the gain is the one-step advantages, collected along the **new** policy's path.

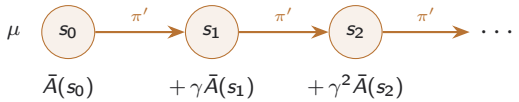
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- **Exact** — the gain is the one-step advantages, collected along the **new** policy's path.
- **Not measurable:** it needs $d_{\pi',\mu}$, the states of a policy we have not run yet.

A New Quantity: the Policy Advantage

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\bar{A}(s)\big]$$

- Lemma 6.1's average, but **on the states π visits** (from μ) — so it is **measurable**.
- One step of policy improvement maximizes it.
- It is exactly the **first-order** change of η_μ in α:

▸ Proof

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') \qquad \Longrightarrow \qquad \Delta\eta_\mu = \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

Small α + positive policy advantage $\Rightarrow$ improvement. Large $\alpha \Rightarrow$ possible degradation. How large can α be?

The Exact Change vs. the Policy Advantage

Apply Lemma 6.1 to the mixture $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi'$: per state, $\mathbb{E}_{a\sim\pi_{\text{new}}} A_\pi(s,a) = \alpha\,\bar{A}(s)$.

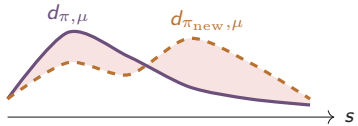
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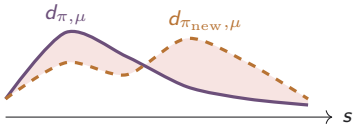
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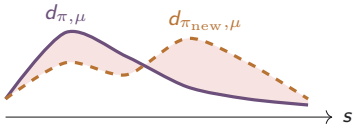
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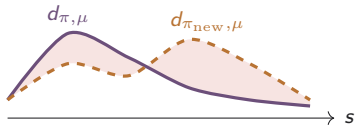
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- $\alpha\mathbb{A}$: the first-order term from the policy-advantage slide.
- Gap: two factors of α — smaller $\bar{A}$ **and** less shift $\Rightarrow O(\alpha^2)$.
- Thm 4.1 = a **lower bound on the gap**.

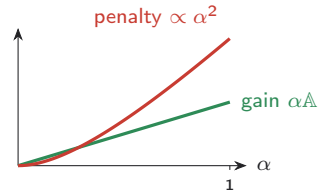
Exact but unmeasurable (Lemma 6.1) = measurable ($\alpha\mathbb{A}$) + gap. We only need to bound the gap.

How Big a Step? The Improvement Bound

Theorem 4.1. With $\varepsilon = \max_s \left|\mathbb{E}_{a\sim\pi'}[A_\pi(s,a)]\right|$, for all $\alpha \in [0,1]$:

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\geq\; \frac{\alpha}{1-\gamma}\left(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\right)$$

- Gain $\alpha\mathbb{A}$: **linear**, measured on old states.
- Penalty: worst case of the state shift, $\propto \alpha^2$.
- Small α: the gain wins $\Rightarrow$ **guaranteed progress**.
- $\alpha = 1$: B&T penalty **plus a gain**. ▸ Compare



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$. Green: improvement guaranteed.

Some small step always helps when $\mathbb{A} > 0$. **Which** α? $\rightarrow$ next slide.

Choosing the Step: Corollary 4.2

1. **Simplify:** lower the bound further, using $|A_\pi| \le R$, so that only the measurable $\mathbb{A}$ is left.

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\Big) \qquad \text{Thm 4.1}$$

$$\;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha R}{1-\gamma}\Big) \;=:\; f(\alpha) \qquad \varepsilon \le R,\;\; 1-\gamma(1-\alpha) \ge 1-\gamma,\;\; \gamma \le 1$$

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2. Maximize: $f(\alpha) = \frac{\mathbb{A}}{1-\gamma}\,\alpha - \frac{2R}{(1-\gamma)^2}\,\alpha^2$ is a downward parabola in α.

$$f'(\alpha) \;=\; \frac{\mathbb{A}}{1-\gamma} - \frac{4R\,\alpha}{(1-\gamma)^2} \;=\; 0 \quad\Longrightarrow\quad \alpha^* \;=\; \frac{(1-\gamma)\mathbb{A}}{4R}$$

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3. Plug in:

$$f(\alpha^*) \;=\; \frac{\mathbb{A}}{1-\gamma}\cdot\frac{(1-\gamma)\mathbb{A}}{4R} \;-\; \frac{2R}{(1-\gamma)^2}\cdot\frac{(1-\gamma)^2\mathbb{A}^2}{16R^2} \;=\; \frac{\mathbb{A}^2}{8R}$$

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

Choosing the Step: Corollary 4.2

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Computable: $\mathbb{A}$ averages over the *old* policy's states (μ-restarts); R is known. Price: $\mathbb{A} \le R$ gives $\alpha \le (1-\gamma)/4$ (≤ 0.0025 at $\gamma = 0.99$), and $\alpha \to 0$ as $\mathbb{A} \to 0$: **tiny steps**.

Conservative Policy Iteration

1. $\pi' \leftarrow G_\varepsilon(\pi, \mu)$ (greedy policy chooser)
2. Estimate $\hat{\mathbb{A}} \approx \mathbb{A}_{\pi,\mu}(\pi')$ to $\pm\frac{\varepsilon}{3}$ with $O\big(\frac{R^2}{\varepsilon^2} \log \frac{R^2}{\delta\varepsilon^2}\big)$ μ-restarts
3. If $\hat{\mathbb{A}} < \frac{2\varepsilon}{3}$: **stop**, return π
4. Else: mix with $\alpha = \frac{(1-\gamma)(\hat{\mathbb{A}} - \varepsilon/3)}{4R}$, go to 1

G_ε: returns π' with $\mathbb{A}_{\pi,\mu}(\pi') \geq \max_{\pi''} \mathbb{A}_{\pi,\mu}(\pi'') - \varepsilon$ — e.g. a regression on advantages with *average* error $\leq \varepsilon/2$.

▸ How?

Theorem 4.4. With prob. $\geq 1 - \delta$, CPI:

i) improves η_μ at **every** update;

ii) stops within $72\, R^2/\varepsilon^2$ calls to G_ε;

iii) returns π with $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$.[1]

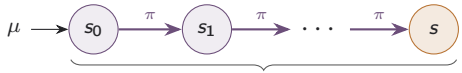
Number of updates does **not** depend on μ or on the state-space size.

[1] $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) \equiv \max_{\pi'} \mathbb{A}_{\pi,\mu}(\pi') = \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\max_a A_\pi(s,a)\big]$: the largest policy advantage any π' can achieve (Def. 4.3). Small $\Rightarrow$ no policy improves much on π under $d_{\pi,\mu}$.

Verifying an Update: Estimating $\mathbb{A}$ From μ-Restarts

Since $V_\pi(s) = \sum_a \pi(a;s)\, Q_\pi(s,a)$, the policy advantage needs only Q_π:

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

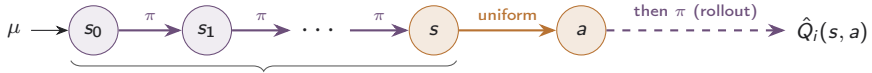


stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

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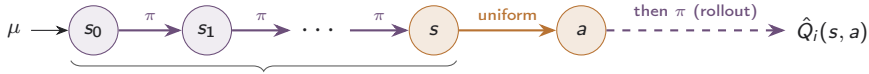


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stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

$$x_i = n_a\, \hat{Q}_i(s,a)\big(\pi'(a;s) - \pi(a;s)\big) \in [-n_a R,\, n_a R], \qquad \hat{\mathbb{A}} = \tfrac{1}{k}\sum_{i=1}^{k} x_i$$

$$\text{Hoeffding:} \quad \Pr\big(|\hat{\mathbb{A}} - \mathbb{A}| > \Delta\big) \;\le\; 2\, e^{-k\Delta^2/(2n_a^2R^2)}$$

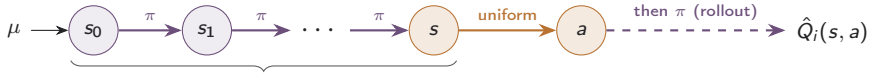
- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, [2]) ▸ IS ▸ Hoeffding

[2] n_a is the number of actions.

Verifying an Update: Estimating $\mathbb{A}$ From μ-Restarts

Since $V_\pi(s) = \sum_a \pi(a;s)\, Q_\pi(s,a)$, the policy advantage needs only Q_π:

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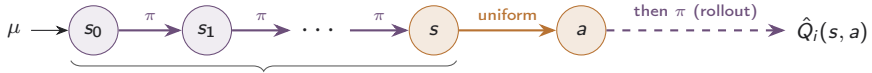
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- Only rollouts of the **current** π — no need to run π_{new}. This is **Q2**.

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stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

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- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, ▸ IS)[2] ▸ Hoeffding
- Only rollouts of the **current** π — no need to run π_{new}. This is **Q2**.
- CPI: $\Delta = \varepsilon/3$, so $\hat{\mathbb{A}} - \frac{\varepsilon}{3} \le \mathbb{A}$ is safe for the stop test and for α.

[2] n_a is the number of actions.

How Good Is the Policy Found?

Theorem 6.2. If $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < \varepsilon$, then for any $\tilde{\mu}$:

$$\eta_{\tilde{\mu}}(\pi^*) - \eta_{\tilde{\mu}}(\pi) \;\le\; \frac{\varepsilon}{1-\gamma} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{d_{\pi,\mu}} \right\|_\infty \;\le\; \frac{\varepsilon}{(1-\gamma)^2} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{\mu} \right\|_\infty$$

- **Covariate-shift bound:** test risk $\le$ density ratio $\times$ train risk.
  - train risk: $\mathrm{OPT}(\mathbb{A}_{\pi,\mu})$, on CPI's data $d_{\pi,\mu}$
  - test: the states π^* visits, $d_{\pi^*,\tilde{\mu}}$
- **Uniform μ caps the ratio**: $d_{\pi,\mu} \ge (1-\gamma)\mu \Rightarrow$ ratio $\le |S|/(1-\gamma)$, for any π^*.

CPI: Answers to the Three Questions

1. **Improving measure:** η_μ improves at every update (Thm. 4.4).
2. **Verification:** estimate the policy advantage $\mathbb{A}$ from samples (Hoeffding).
3. **Performance:** within $O(R^2/\varepsilon^2)$ updates, close to optimal up to the mismatch factor (Thm. 6.2).

Kept going forward

- Surrogate on the old state distribution
- Improvement lower bound = gain − penalty
- Small, controlled steps

Limitations

- Only **mixture** policies
- Needs a restart distribution
- Theoretical step sizes are tiny

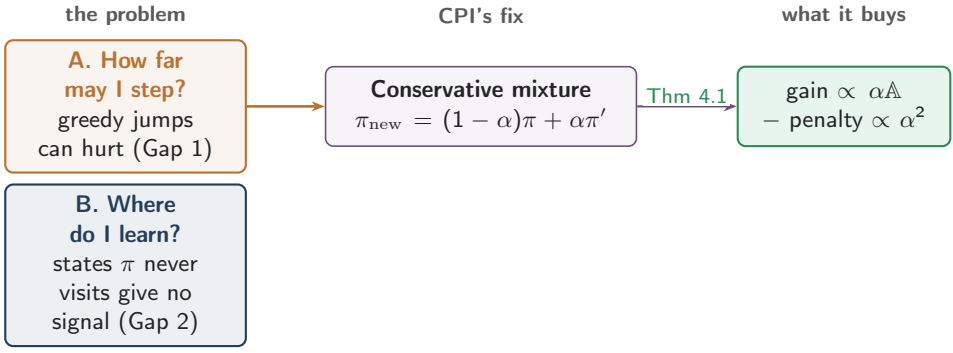
Like non-convex SGD, but Thm. 6.2 turns "stationary" into "near-optimal".

▸ CPI vs. SGD

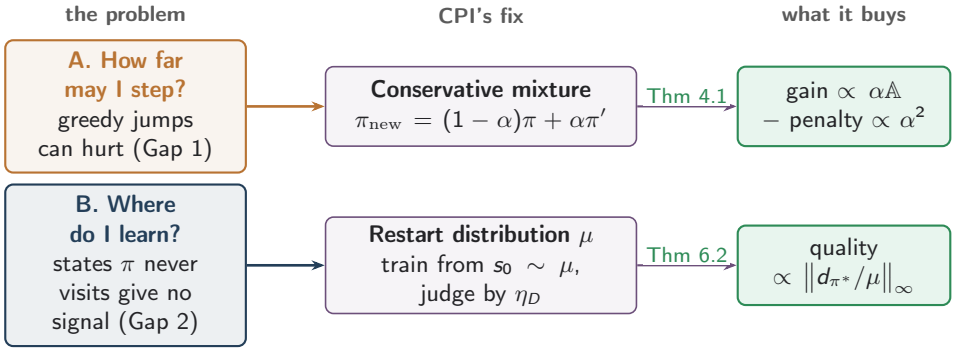
Recap: Two Problems, Two Fixes

the problem	CPI's fix	what it buys
A. How far may I step? greedy jumps can hurt (Gap 1)		
B. Where do I learn? states π never visits give no signal (Gap 2)		

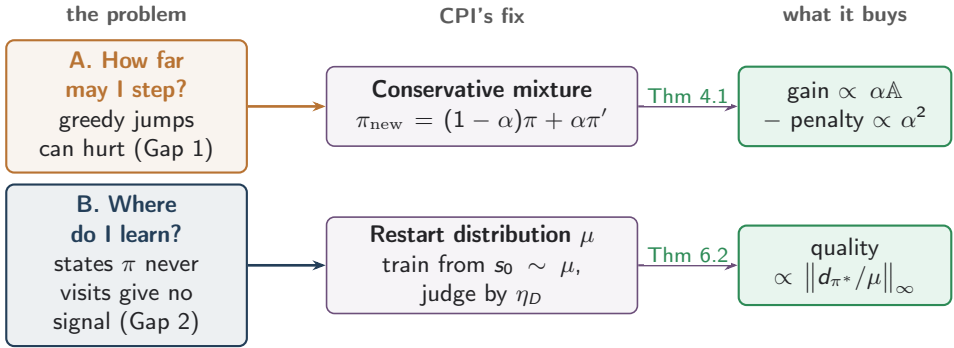
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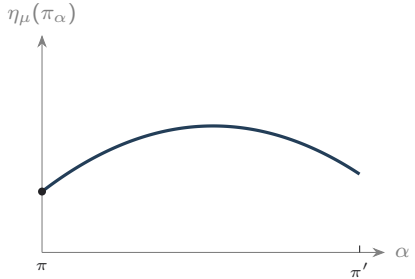


Two **separate** axes: **update size** (A) and **state coverage** (B). We follow both through TRPO, GAE and PPO.

The Policy Advantage Is a Gradient

The mixtures are a one-parameter policy class

$$\pi_\alpha = (1-\alpha)\,\pi + \alpha\,\pi', \qquad \partial_\alpha \pi_\alpha(a;s) = \pi'(a;s) - \pi(a;s)$$



Moving from π toward π' = walking along one line in policy space.

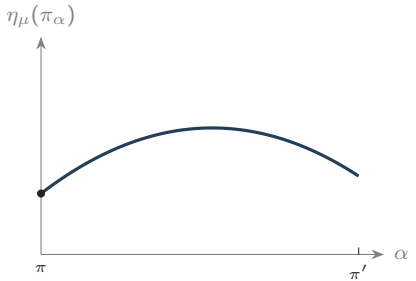
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Policy gradient theorem with $\theta := \alpha$, at $\alpha = 0$

$$\partial_\alpha \eta_\mu\big|_0 = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\textstyle\sum_a \big(\pi' - \pi\big)(a;s)\,Q_\pi(s,a)\Big]$$



The theorem never asks what the parameter *means*.

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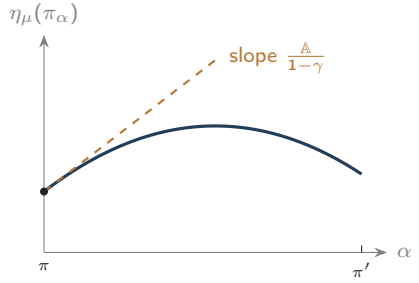
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$\sum_a \pi\, Q_\pi = V_\pi$ and $\sum_a \pi' = 1$ turn Q_π into A_π

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$\mathbb{A}$ is the **slope** of η_μ in the direction of π'.

The Policy Advantage Is a Gradient

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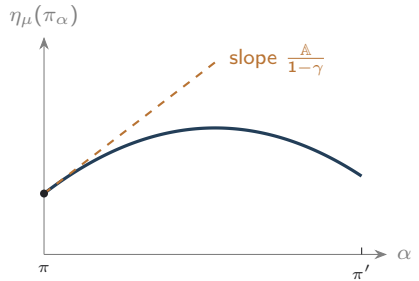
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$\mathbb{A}$ is the **slope** of η_μ in the direction of π'.

α plays no special role — it is just a coordinate. Any π_θ gives the same first-order picture $\Rightarrow$ **TRPO**.

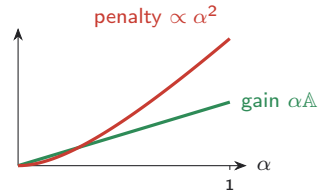
▸ Full proof

How Big a Step? The Improvement Bound

Theorem 4.1. With $\varepsilon = \max_s \left|\mathbb{E}_{a\sim\pi'}[A_\pi(s,a)]\right|$, for all $\alpha \in [0,1]$:

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\geq\; \frac{\alpha}{1-\gamma}\left(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\right)$$

- Gain $\alpha\mathbb{A}$: **linear**, measured on old states.
- Penalty: worst case of the state shift, $\propto \alpha^2$.
- Small α: the gain wins $\Rightarrow$ **guaranteed progress**.
- $\alpha = 1$: B&T penalty **plus a gain**. ▸ Compare



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$. Green: improvement guaranteed.

Some small step always helps when $\mathbb{A} > 0$. **Which** α? $\rightarrow$ next slide.

Choosing the Step: Corollary 4.2

1. **Simplify:** lower the bound further, using $|A_\pi| \le R$, so that only the measurable $\mathbb{A}$ is left.

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\Big) \qquad \text{Thm 4.1}$$

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$$\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha R}{1-\gamma}\Big) \;=:\; f(\alpha) \quad \varepsilon \le R,\;\; 1-\gamma(1-\alpha) \ge 1-\gamma,\;\; \gamma \le 1$$

2. Maximize: $f(\alpha) = \frac{\mathbb{A}}{1-\gamma}\,\alpha - \frac{2R}{(1-\gamma)^2}\,\alpha^2$ is a downward parabola in α.

$$f'(\alpha) \;=\; \frac{\mathbb{A}}{1-\gamma} - \frac{4R\,\alpha}{(1-\gamma)^2} \;=\; 0 \quad\Longrightarrow\quad \alpha^* \;=\; \frac{(1-\gamma)\mathbb{A}}{4R}$$

Choosing the Step: Corollary 4.2

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3. Plug in:

$$f(\alpha^*) \;=\; \frac{\mathbb{A}}{1-\gamma}\cdot\frac{(1-\gamma)\mathbb{A}}{4R} \;-\; \frac{2R}{(1-\gamma)^2}\cdot\frac{(1-\gamma)^2\mathbb{A}^2}{16R^2} \;=\; \frac{\mathbb{A}^2}{8R}$$

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

Computable: $\mathbb{A}$ averages over the *old* policy's states (μ-restarts); R is known. Price: $\mathbb{A} \le R$ gives $\alpha \le (1-\gamma)/4$ (≤ 0.0025 at $\gamma = 0.99$), and $\alpha \to 0$ as $\mathbb{A} \to 0$: **tiny steps**.

Conservative Policy Iteration

1. $\pi' \leftarrow G_\varepsilon(\pi, \mu)$ (greedy policy chooser)
2. Estimate $\hat{\mathbb{A}} \approx \mathbb{A}_{\pi,\mu}(\pi')$ to $\pm\frac{\varepsilon}{3}$ with $O\big(\frac{R^2}{\varepsilon^2} \log \frac{R^2}{\delta\varepsilon^2}\big)$ μ-restarts
3. If $\hat{\mathbb{A}} < \frac{2\varepsilon}{3}$: **stop**, return π
4. Else: mix with $\alpha = \frac{(1-\gamma)(\hat{\mathbb{A}} - \varepsilon/3)}{4R}$, go to 1

G_ε: returns π' with $\mathbb{A}_{\pi,\mu}(\pi') \geq \max_{\pi''} \mathbb{A}_{\pi,\mu}(\pi'') - \varepsilon$ — e.g. a regression on advantages with *average* error $\leq \varepsilon/2$.

▸ How?

Theorem 4.4. With prob. $\geq 1 - \delta$, CPI:

i) improves η_μ at **every** update;

ii) stops within $72\, R^2/\varepsilon^2$ calls to G_ε;

iii) returns π with $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$.[1]

Number of updates does **not** depend on μ or on the state-space size.

[1] $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) \equiv \max_{\pi'} \mathbb{A}_{\pi,\mu}(\pi') = \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\max_a A_\pi(s,a)\big]$: the largest policy advantage any π' can achieve (Def. 4.3). Small $\Rightarrow$ no policy improves much on π under $d_{\pi,\mu}$.

CPI: Answers to the Three Questions

1. **Improving measure:** η_μ improves at every update (Thm. 4.4).
2. **Verification:** estimate the policy advantage $\mathbb{A}$ from samples (Hoeffding).
3. **Performance:** within $O(R^2/\varepsilon^2)$ updates, close to optimal up to the mismatch factor (Thm. 6.2).

Kept going forward

- Surrogate on the old state distribution
- Improvement lower bound = gain − penalty
- Small, controlled steps

Limitations

- Only **mixture** policies
- Needs a restart distribution
- Theoretical step sizes are tiny

Like non-convex SGD, but Thm. 6.2 turns "stationary" into "near-optimal".

▸ CPI vs. SGD

Trust Region Policy Optimization

Schulman, Levine, Moritz, Jordan & Abbeel
ICML 2015

Introduces **TRPO**: CPI's guarantee for **any** stochastic policy π_θ

Setting & Assumptions

An MDP $(\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$:

- **Finite** $\mathcal{S}, \mathcal{A}$ in the proofs — also holds for **continuous** (conditions later).
- **Discounted**: $\gamma \in (0, 1)$, reward $r(s)$.
- **Unknown dynamics**: sample trajectories only.
- **Start distribution** ρ_0 (CPI's D) — **no restart** μ.
- **Unnormalized**: $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$, $s_0 \sim \rho_0$.
- **Any** differentiable stochastic policy π_θ (e.g. a neural net).

From CPI to TRPO: What Changes

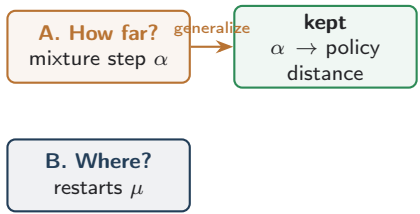
	CPI	TRPO
policies	mixtures π_α	any π_θ (neural net)
new candidate	greedy chooser G_ε	a step in θ
"how far"	weight α	distance $D(\pi_{\text{old}}, \pi_{\text{new}})$
start states	D, trained from μ	ρ_0 only
scale	normalized by $1-\gamma$	unnormalized

A. How far?
mixture step α

B. Where?
restarts μ

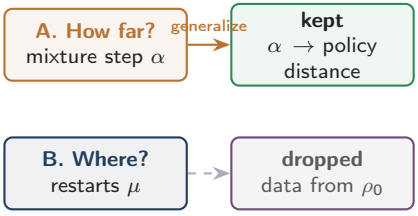
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start states	D, trained from μ	ρ_0 only
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From CPI to TRPO: What Changes

	CPI	TRPO
policies	mixtures π_α	any π_θ (neural net)
new candidate	greedy chooser G_ε	a step in θ
"how far"	weight α	distance $D(\pi_{\text{old}}, \pi_{\text{new}})$
start states	D, trained from μ	ρ_0 only
scale	normalized by $1-\gamma$	unnormalized



TRPO answers **A** for neural-network policies. It does **not** carry over CPI's answer to **B** — we come back to this.

TRPO: Starting From the Same Identity

Unnormalized return $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$ and visitation $\rho_\pi(s) = \sum_t \gamma^t P(s_t = s)$.

$$\eta(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a) \qquad \text{(performance difference)}$$

$\rho_{\tilde{\pi}}$ depends on $\tilde{\pi}$ in a complicated way $\Rightarrow$ replace it by the **old** visitation:

$$L_\pi(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_\pi(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a)$$

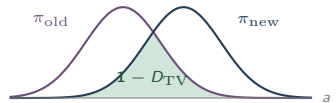
- Matches η to first order: $L_{\pi_{\theta_0}}(\pi_{\theta_0}) = \eta(\pi_{\theta_0})$ and equal gradients at θ_0.
- Same idea as CPI's policy advantage — but says nothing about **how big** a step is safe.

From Mixtures to Any Stochastic Policy

CPI's bound only holds for mixtures. TRPO replaces the mixing weight α by a **distance** between policies:

$$D_{\mathrm{TV}}(p \,\|\, q) = \tfrac{1}{2} \sum_a \left| p(a) - q(a) \right|$$

$$\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}}) = \max_s D_{\mathrm{TV}}\big(\pi_{\mathrm{old}}(\cdot \mid s) \,\|\, \pi_{\mathrm{new}}(\cdot \mid s)\big)$$



Shared mass: both pick the **same** action w.p. $1 - D_{\mathrm{TV}}$.

Theorem 1.

$$\eta(\pi_{\mathrm{new}}) \;\geq\; L_{\pi_{\mathrm{old}}}(\pi_{\mathrm{new}}) - \frac{4\epsilon\gamma}{(1-\gamma)^2}\,\alpha^2, \qquad \epsilon = \max_{s,a} |A_\pi(s,a)|$$

- Proof idea: **couple** the two policies — they agree w.p. $1 - \alpha$, so the new policy "acts like a mixture".
- Different ϵ: max over (s, a), not CPI's max over s of an expectation.

▸ Continuous S, A?

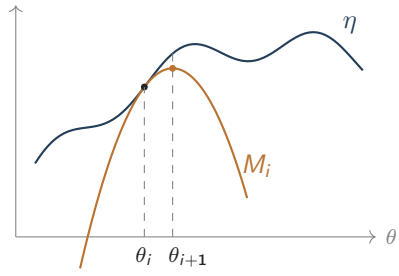
From TV to KL: a Minorize–Maximize Algorithm

KL divergence: $D_{\mathrm{KL}}(p\|q) = \sum_a p(a) \log \frac{p(a)}{q(a)} \geq 0$, smooth in θ.

Use $D_{\mathrm{TV}}(p\|q)^2 \leq D_{\mathrm{KL}}(p\|q)$:

$$\eta(\tilde{\pi}) \geq \underbrace{L_\pi(\tilde{\pi}) - C\, D_{\mathrm{KL}}^{\max}(\pi, \tilde{\pi})}_{M_i(\tilde{\pi})}, \quad C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$$

- $M_i \leq \eta$ everywhere, with equality at π_i.
- Maximize $M_i \Rightarrow$ $\eta(\pi_{i+1}) - \eta(\pi_i) \geq M_i(\pi_{i+1}) - M_i(\pi_i) \geq 0$.
- **Monotone improvement** (with exact advantages).



This is the formulation **the theory covers**. Next: five simplifications to make it run.

From Penalty to Constraint: the Trust Region

Theory (penalty)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) - C\, D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta)$$

With the theoretical C, steps are **tiny**; and a good C is hard to choose robustly.

Practice (constraint)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta)$$

$$\text{s.t.} \;\; D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \le \delta$$

δ is in policy space, so it transfers across problems ($\delta = 0.01$ in all experiments).

Why needed

$C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$: at $\gamma = 0.99$, $C \approx 4 \cdot 10^4\, \epsilon \Rightarrow$ tiny steps.

What changes

A hard limit δ on the KL replaces the penalty C: we optimize L where we *trust* it.

What it costs

The guarantee becomes a heuristic; δ is a hyperparameter.

From Max KL to Mean KL

1. penalty → constraint → **2. max → mean KL** → 3. samples + IS → 4. linear / quadratic → 5. CG + line search

Replace the max over states by the average under the old visitation distribution:

$$\bar{D}_{\mathrm{KL}}^{\rho}(\theta_1, \theta_2) = \mathbb{E}_{s\sim\rho}\Big[D_{\mathrm{KL}}\big(\pi_{\theta_1}(\cdot \mid s) \,\|\, \pi_{\theta_2}(\cdot \mid s)\big)\Big]$$

$$\max_{\theta} \; L_{\theta_{\mathrm{old}}}(\theta) \quad \text{s.t.} \quad \bar{D}_{\mathrm{KL}}^{\rho_{\theta_{\mathrm{old}}}}(\theta_{\mathrm{old}}, \theta) \le \delta$$

Why needed

$D_{\mathrm{KL}}^{\max}$ is one constraint **per state**, also at states we never sample.

What changes

An **expectation** over visited states $\Rightarrow$ estimated from the same rollouts.

What it costs

Rarely visited states may change a lot. Empirically like max KL (results at the end).

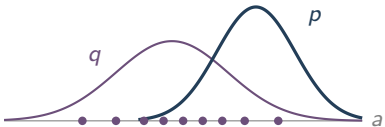
Tool: Importance Sampling

Monte Carlo: an expectation $\approx$ an average of samples

$$\mathbb{E}_{a\sim p}[f] \approx \tfrac{1}{n}\textstyle\sum_i f(a_i), \quad a_i \sim p$$

Samples only from q? Weight $w = p/q$ ($q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$



Samples come from q, but we want an average under p.

Tool: Importance Sampling

Monte Carlo: an expectation $\approx$ an average of samples

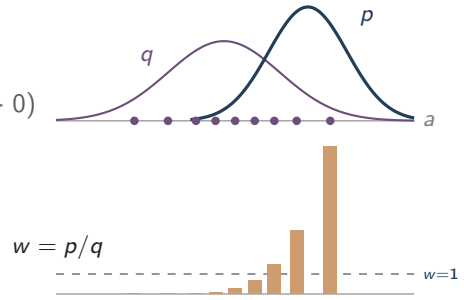
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$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$



p far from q: one rare sample carries a huge weight $\Rightarrow$ high variance.

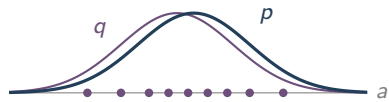
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Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$

In TRPO (single path), at each state s

$$p = \pi_\theta(\cdot \mid s), \quad q = \pi_{\theta_{\text{old}}}(\cdot \mid s)$$

$$w = \frac{\pi_\theta(a \mid s)}{\pi_{\theta_{\text{old}}}(a \mid s)}$$

$w = p/q$



Trust region keeps $\pi_\theta \approx \pi_{\theta_{\text{old}}} \Rightarrow$ all $w \approx 1$.

Only actions are reweighted — states stay $s \sim \rho_{\theta_{\text{old}}}$.

Turning Sums Into Samples

1. penalty → constraint → 2. max → mean KL → **3. samples + IS** → 4. linear / quadratic → 5. CG + line search

$$\max_\theta \sum_s \rho_{\theta_{\text{old}}}(s) \sum_a \pi_\theta(a \mid s)\, A_{\theta_{\text{old}}}(s,a)$$

1. $\sum_s \rho_{\theta_{\text{old}}}(s)[\cdot] \;\to\; \frac{1}{1-\gamma}\mathbb{E}_{s\sim\rho_{\theta_{\text{old}}}}[\cdot]$ (visited states)
2. $A_{\theta_{\text{old}}} \;\to\; Q_{\theta_{\text{old}}}$ (changes the objective by a constant)
3. $\sum_a \;\to$ **importance sampling** with a sampling distribution q (previous slide)

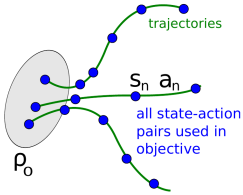
$$\max_\theta \; \mathbb{E}_{s\sim\rho_{\theta_{\text{old}}},\, a\sim q}\left[\frac{\pi_\theta(a \mid s)}{q(a \mid s)}\, Q_{\theta_{\text{old}}}(s,a)\right]$$
$$\text{s.t.} \;\; \mathbb{E}_{s\sim\rho_{\theta_{\text{old}}}}\left[D_{\text{KL}}(\pi_{\theta_{\text{old}}}(\cdot \mid s) \,\|\, \pi_\theta(\cdot \mid s))\right] \le \delta$$

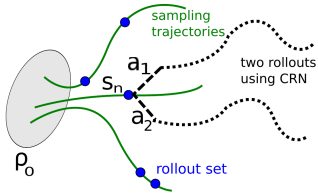
Why needed
Sums over all s and a need the model; continuous a has no sum at

What changes
Averages over rollouts of π_{old}; ratio π_θ/q; $\hat{Q}$ from returns.

What it costs
Noise (variance) in L and its gradient; states are still only

Two Sampling Schemes





Single path

- Roll out $\pi_{\theta_{\text{old}}}$; use **every** (s_n, a_n).
- $q = \pi_{\theta_{\text{old}}}$; $\hat{Q}$ = discounted return-to-go.
- **No state resets** — works on real systems.

Vine

- From each s_n in the **rollout set**: try K actions $a_k \sim q$, short rollout each (CRN).
- **Lower variance** $\hat{Q}$.
- Needs **resets** — but only to states reached from ρ_0: **not** CPI's μ.

Solving the Trust-Region Problem: Search Direction

1. penalty → constraint → 2. max → mean KL → 3. samples + IS → **4. linear / quadratic** → 5. CG + line search

Approximate around θ_{old} — **linear** objective, **quadratic** constraint:

$$L(\theta) \approx L(\theta_{\text{old}}) + g^\top(\theta - \theta_{\text{old}}), \qquad \bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \approx \tfrac{1}{2}(\theta - \theta_{\text{old}})^\top A\,(\theta - \theta_{\text{old}})$$

- $A_{ij} = \frac{\partial^2}{\partial\theta_i \partial\theta_j}\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta)$: the **curvature of the KL**, a.k.a. the **Fisher information matrix**.
- Solution direction: $s \propto A^{-1}g$ — the **natural gradient**: steepest ascent when distance is measured in KL, not in $\|\theta\|$.

Why needed
L and the KL are nonlinear in θ; an exact constrained solve per batch is too expensive.

What changes
First/second-order Taylor models $\Rightarrow$ a closed-form direction $A^{-1}g$.

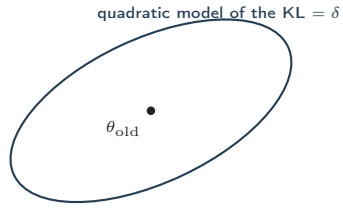
What it costs
Valid only locally — the step must be checked (next slide).

Computing the Step: Conjugate Gradient + Line Search

1. penalty → constraint → 2. max → mean KL → 3. samples + IS → 4. linear / quadratic → **5. CG + line search**

Direction: solve $A\,s = g$ by **conjugate gradient**

only needs products $y \mapsto Ay$ (autodiff), $k = 10$ iterations



A has $|\theta|^2$ entries — never formed.

Why needed
A cannot even be stored for a neural net; the quadratic model is only local.

What changes
CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs
Second order and **one step per batch** — PPO's starting point.

Computing the Step: Conjugate Gradient + Line Search

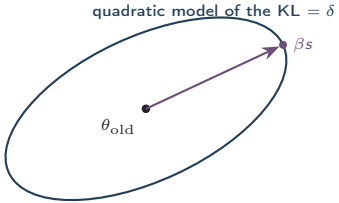
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Largest step that reaches the KL boundary

$$\tfrac{1}{2}(\beta s)^\top A(\beta s) = \delta \;\Rightarrow\; \beta = \sqrt{2\delta / s^\top A s}$$



Step to the edge of the *model*'s trust region.

Why needed
A cannot even be stored for a neural net; the quadratic model is only local.

What changes
CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs
Second order and **one step per batch** — PPO's starting point.

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Direction: solve $A\,s = g$ by **conjugate gradient**

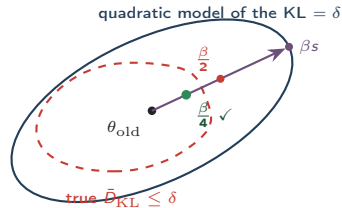
only needs products $y \mapsto Ay$ (autodiff), $k = 10$ iterations

Largest step that reaches the KL boundary

$$\tfrac{1}{2}(\beta s)^\top A(\beta s) = \delta \;\Rightarrow\; \beta = \sqrt{2\delta / s^\top A s}$$

Backtrack $\beta,\ \beta/2,\ \beta/4, \ldots$ until

$L_{\theta_{\text{old}}}(\theta)$ improves and $\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \le \delta$



The model is only local: shrink until the **true** checks pass.

Why needed

A cannot even be stored for a neural net; the quadratic model is only local.

What changes

CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs

Second order and **one step per batch** — PPO's starting point.

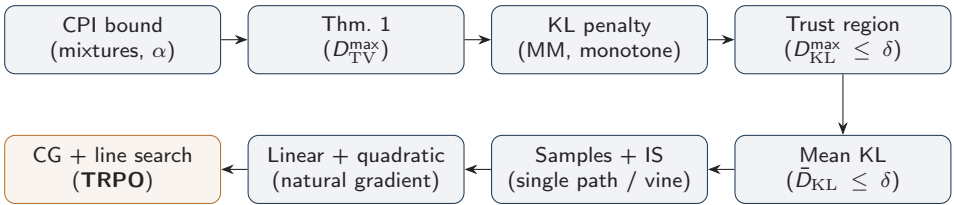
The TRPO Algorithm

Repeat:

1. Collect (s, a) pairs with Monte Carlo $\hat{Q}$ estimates (single path or vine).
2. Build the sample estimates of the surrogate L and the mean-KL constraint.
3. Compute $g = \nabla_\theta L$; solve $As = g$ with conjugate gradient (Fisher-vector products).
4. Step size $\beta = \sqrt{2\delta / s^\top As}$; backtracking line search on L and the KL.
5. $\theta_{\text{old}} \leftarrow \theta_{\text{old}} + \beta s$.

Cost per update: only slightly more than computing the gradient itself.

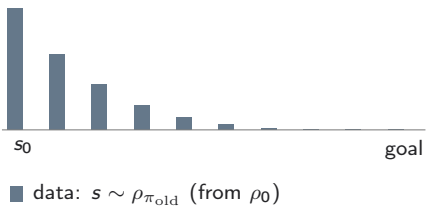
Recap: From Theory to Practice



- First three boxes: **theory** (monotone improvement). From the trust region on: **practice** — each arrow trades a guarantee for practicality.
- What's left: noisy $\hat{Q}$ (GAE) and a heavy second-order solver (PPO).

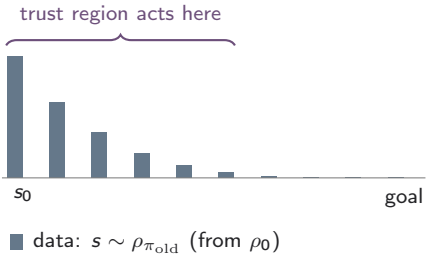
▸ Related algorithms

What TRPO Does Not Fix: Where the Data Come From



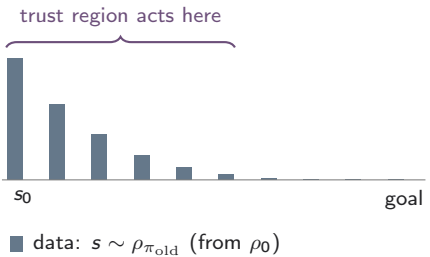
- **Data:** $s_0 \sim \rho_0$, then π_{old} — CPI's μ is gone.

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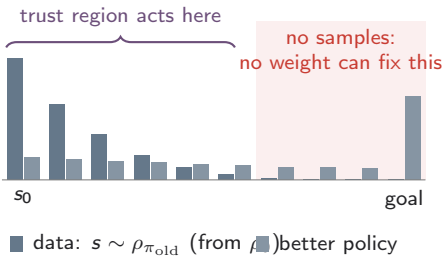
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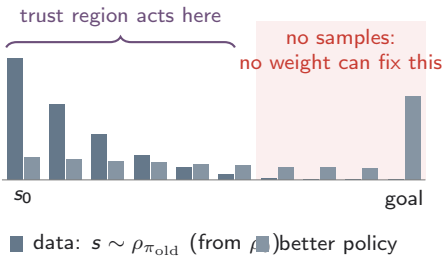
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- **Trust region:** limits how much π changes **on the states we have**.
- **Importance sampling:** reweights **actions**, never states.
- **Vine:** resets only to states already reached from ρ_0.

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Trust region $\neq$ exploration. Lane **B** (CPI's Gap 2) stays open in TRPO, GAE and PPO.

▸ Gap 2, Ex. 1

GAE and PPO — Main Sources



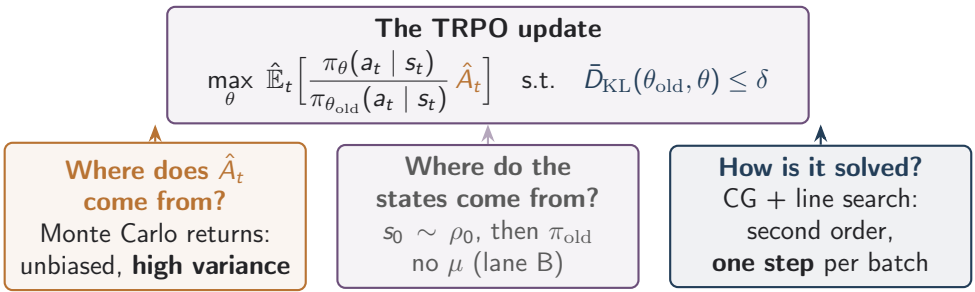
Published parallel to TRPO, this introduced the exponentially-weighted advantage estimator used inside almost all modern PPO and TRPO implementations to balance variance and bias.

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l (r_{t+l} + \gamma V(s_{t+l+1}) - V(s_{t+l}))$$

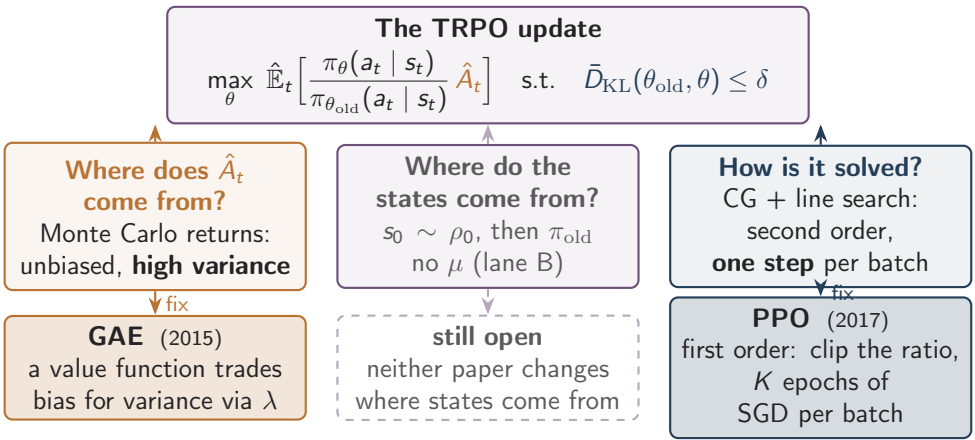


Simplified TRPO by replacing the expensive KL penalty with a clipped surrogate objective function. This became the industry standard for Reinforcement Learning from Human Feedback (RLHF).

Where TRPO Leaves Us



Where TRPO Leaves Us



High-Dimensional Continuous Control Using Generalized Advantage Estimation

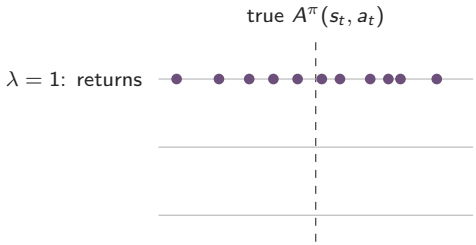
Schulman, Moritz, Levine, Jordan & Abbeel
ICLR 2016 (arXiv 2015)

Introduces **GAE**(γ, λ): TD(λ) for advantages

Setting & Assumptions

- Same MDP as TRPO: start distribution ρ_0, unknown dynamics, reward $r_t = r(s_t, a_t, s_{t+1})$.
- **Undiscounted** goal: maximize $\mathbb{E}\big[\sum_t r_t\big]$ (assumed finite).
- γ is **not** part of the problem: an algorithm knob that trades bias for variance.
- Any policy-gradient method — here TRPO for the policy, a trust region for V_ϕ too.
- 3D robots, neural nets with $> 10^4$ parameters, torques from raw kinematics.

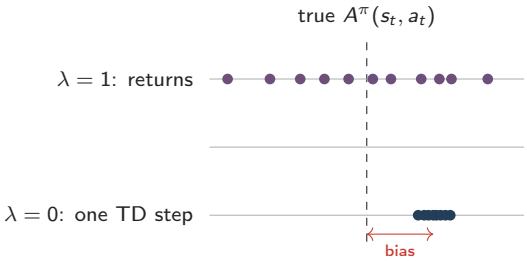
What GAE Fixes — and What It Does Not



Each dot: one estimate $\hat{A}_t$ of the same (s_t, a_t) (schematic).

- TRPO multiplies each ratio by $\hat{A}_t$. From one rollout: **unbiased**, but **very noisy**.

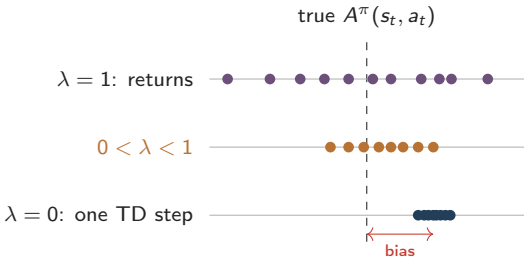
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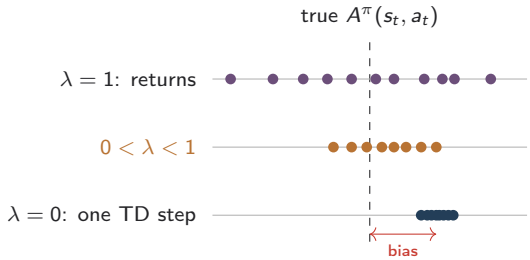
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Fixes: noisy update directions $\Rightarrow$ fewer samples per step.

Not an exploration knob: λ changes how we *score* sampled actions, not *which states* we visit.

Which Weight Multiplies the Score Function?

$$g = \mathbb{E}\Big[\sum_{t\geq 0} \Psi_t \, \nabla_\theta \log \pi_\theta(a_t \mid s_t)\Big]$$

Same expectation for each choice of Ψ_t — **different variance**:

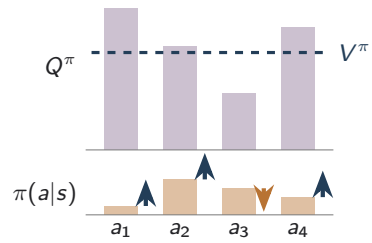
1	$\sum_{t'\geq 0} r_{t'}$	total return
2	$\sum_{t'\geq t} r_{t'}$	reward after a_t
3	$\sum_{t'\geq t} r_{t'} - b(s_t)$	with a baseline
4	$Q^\pi(s_t, a_t)$	action value
5	$A^\pi(s_t, a_t)$	advantage
6	$r_t + V^\pi(s_{t+1}) - V^\pi(s_t)$	TD residual

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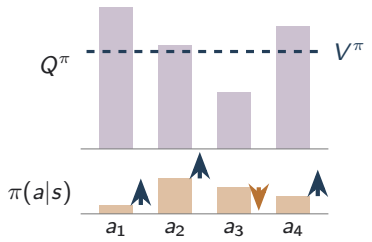
Step along $A \nabla \log \pi$: raise $\pi(a|s)$ **iff** a beats the average V^π.

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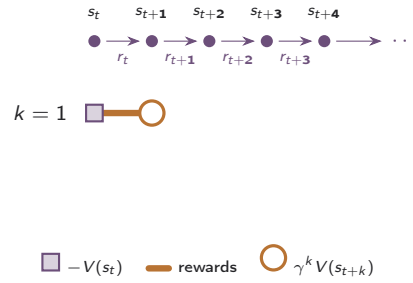
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Discount as a knob: $A^{\pi,\gamma}$ ignores effects $\gg \frac{1}{1-\gamma}$ steps later $\Rightarrow$ less variance, **some bias**.

GAE: TD(λ) for Advantages

TD residual: one reward, then V

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$



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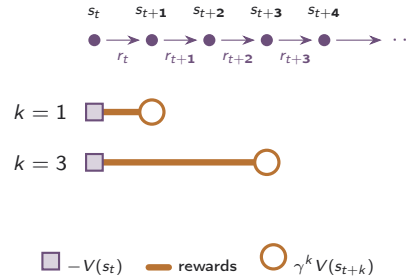
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$$\hat{A}_t^{(k)} = -V(s_t) + \sum_{l=0}^{k-1} \gamma^l r_{t+l} + \gamma^k V(s_{t+k})$$



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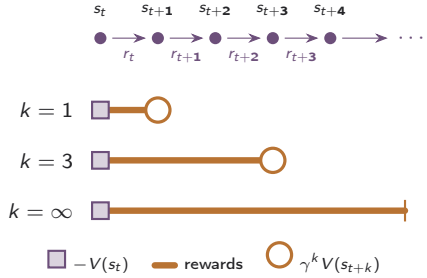
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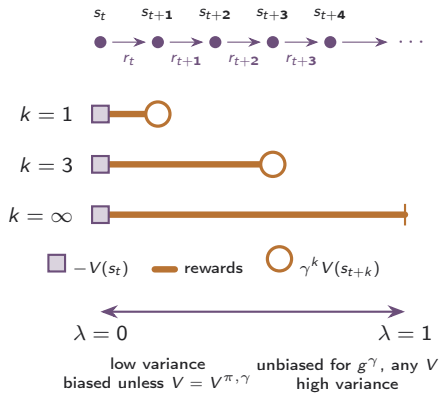
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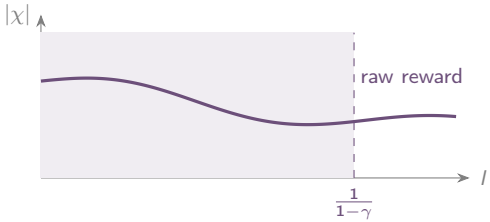
$$\hat{A}_t^{(\infty)} = \sum_{l \geq 0} \gamma^l r_{t+l} - V(s_t)$$

GAE: average all k with weights $(1-\lambda)\lambda^{k-1}$

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Two Knobs, Two Kinds of Bias



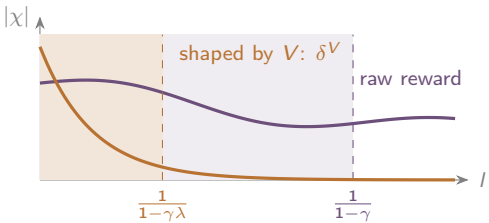
$\chi(l; s, a) = \mathbb{E}[r_{t+l} \mid s, a] - \mathbb{E}[r_{t+l} \mid s]$: how long a_t keeps mattering

γ: cut the horizon

Drops effects after $\approx \frac{1}{1-\gamma}$ steps.

Biased **even with the exact** V.

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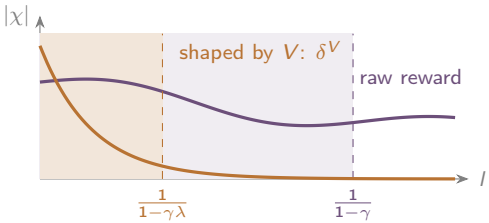
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λ: shape, then cut again

δ^V is the reward shaped by $\Phi = V$: a good V pulls the effect of a_t to $l \approx 0$.

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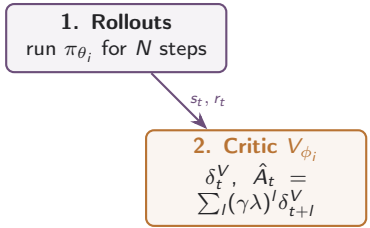
So the best λ is **lower** than the best γ (numbers at the end).

Actor–Critic: Policy and Value Networks

1. Rollouts
run π_{θ_i} for N steps

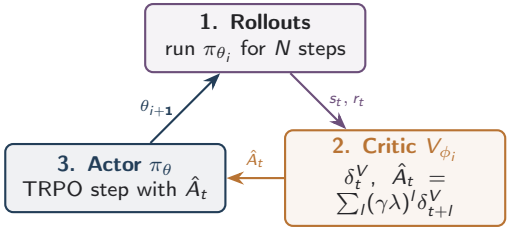
- **Actor**: the policy network, updated by TRPO.
- **Critic**: a second network $V_\phi(s) \approx V^{\pi,\gamma}(s)$, fitted by regression to returns.

Actor–Critic: Policy and Value Networks



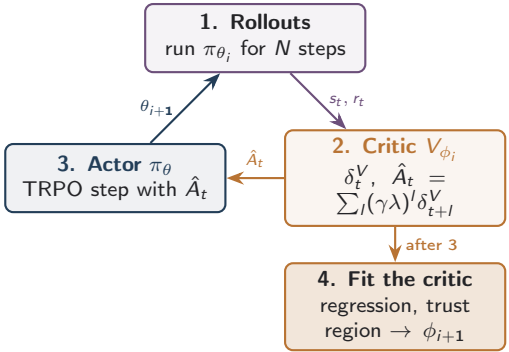
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Actor–Critic: Policy and Value Networks



Order matters: fit V *after* computing $\hat{A}_t$ — an overfit V makes every $\delta_t^V = 0$.

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Value fit with a trust region

$$\min_\phi \sum_n \left\| V_\phi(s_n) - \hat{V}_n \right\|^2$$

$$\text{s.t.} \quad \frac{1}{N} \sum_n \frac{\| V_\phi(s_n) - V_{\phi_{\text{old}}}(s_n) \|^2}{2\sigma^2} \leq \epsilon$$

= a KL trust region for a Gaussian with mean $V_\phi(s)$; solved with CG, like TRPO.

Proximal Policy Optimization Algorithms

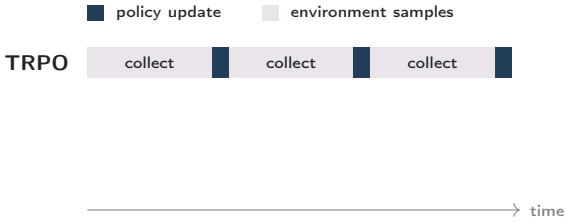
Schulman, Wolski, Dhariwal, Radford & Klimov
arXiv 2017 (OpenAI)

Introduces **PPO**: TRPO's reliability with only **first-order** SGD

Setting & Motivation

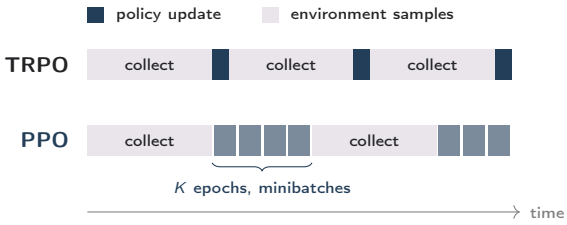
- Same setting as TRPO; advantages $\hat{A}_t$ from (truncated) **GAE**.
- **TRPO**: complicated; no dropout, no parameter sharing between policy and value.
- **Vanilla PG**: several steps on the same batch $\Rightarrow$ destructively large updates.
- **Goal**: reuse each batch for many SGD steps, without leaving the trust region.
- Tested on MuJoCo and 49 Atari games.

PPO's Goal: Many SGD Steps per Batch



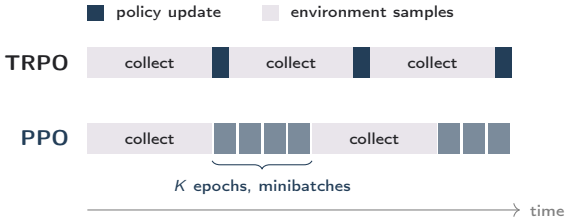
TRPO: one (expensive) step, then the batch is thrown away.

PPO's Goal: Many SGD Steps per Batch

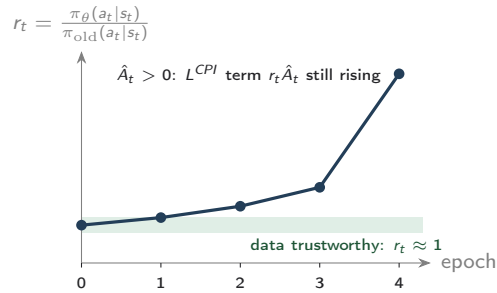


PPO: squeeze the same batch with cheap first-order steps.

PPO's Goal: Many SGD Steps per Batch

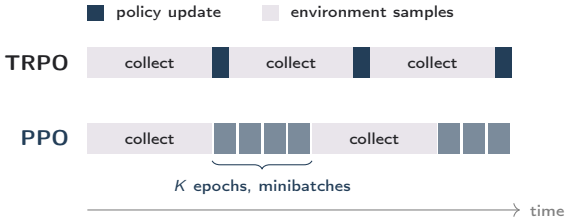


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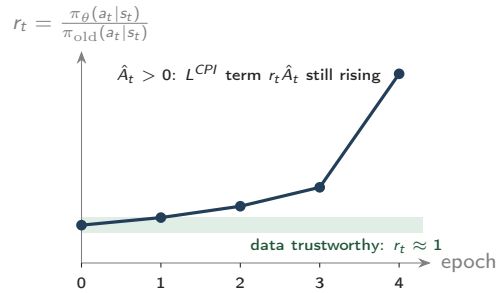


One sample, $\hat{A}_t > 0$: **nothing says stop**.

PPO's Goal: Many SGD Steps per Batch



PPO: squeeze the same batch with cheap first-order steps.



One sample, $\hat{A}_t > 0$: **nothing says stop**.

The data came from π_{old}: the surrogate is trustworthy only near $r_t = 1$. Unchecked, repeated steps give **destructively large** updates.

Back to the CPI Surrogate

Ratio = TRPO's importance weight: $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \quad r_t(\theta_{\text{old}}) = 1$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t\left[r_t(\theta)\,\hat{A}_t\right] \qquad \text{("CPI": Kakade \& Langford)}$$

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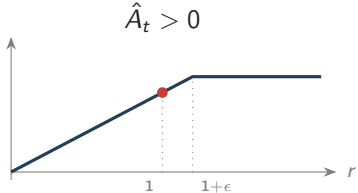
Constraint (TRPO)
s.t. $\hat{\mathbb{E}}_t[D_{\text{KL}}] \le \delta$
second-order solver

Penalty (baseline)
$-\beta\,\hat{\mathbb{E}}_t[D_{\text{KL}}]$
no single β works $\rightarrow$ adapt it

Clip ("PPO")
no gain once $|r_t - 1| > \epsilon$
first order, next slide

The Clipped Surrogate Objective

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min\left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\right)\Big], \qquad \epsilon = 0.2$$

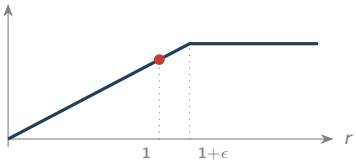


Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

The Clipped Surrogate Objective

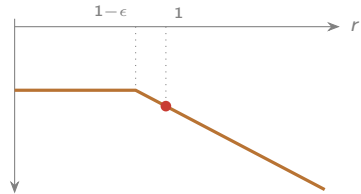
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$\hat{A}_t > 0$



Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

$\hat{A}_t < 0$

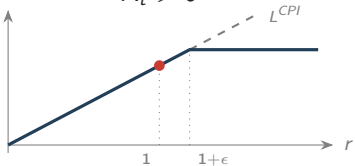


Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.

The Clipped Surrogate Objective

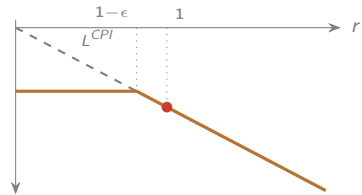
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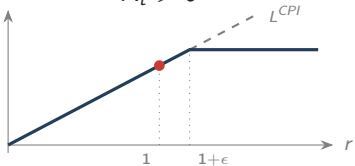
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min $\Rightarrow$ $L^{CLIP} \leq L^{CPI}$: clipping removes **gains**, never hides **losses** — a pessimistic bound.

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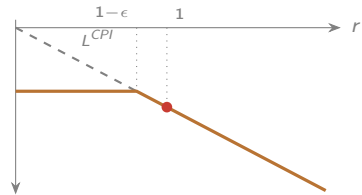
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$\hat{A}_t > 0$



Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.
Wrong way ($r < 1-\epsilon$): not clipped, still costs.

$\hat{A}_t < 0$



Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.
Wrong way ($r > 1+\epsilon$): not clipped, loss keeps growing.

min $\Rightarrow$ $L^{CLIP} \leq L^{CPI}$: clipping removes **gains**, never hides **losses** — a pessimistic bound.
Moving the wrong way keeps its gradient: the update can always pull r_t **back** toward 1.

What Clipping Replaces

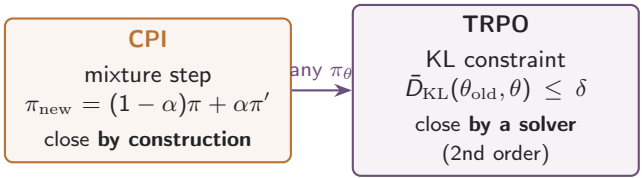
CPI

mixture step

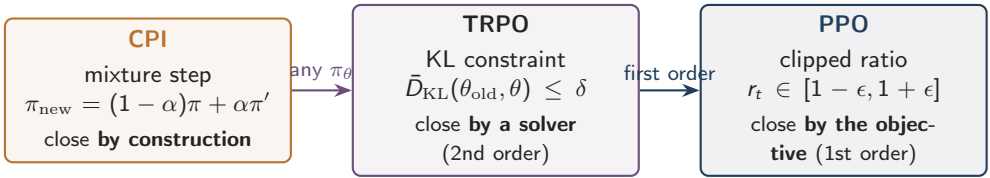
$\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi'$

close **by construction**

What Clipping Replaces

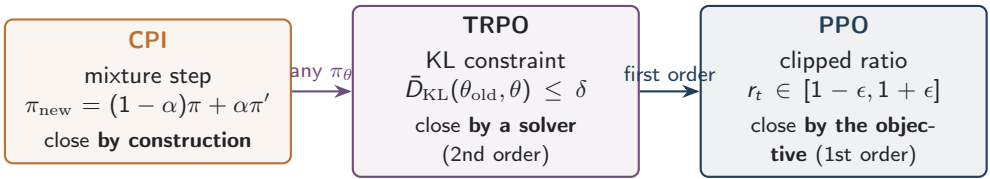


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Proximal: keep π_θ near π_{old} — with plain minibatch SGD/Adam, several epochs per batch.

What Clipping Replaces



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Clipping removes the **incentive** to move a ratio further — it does not **enforce** a limit. No KL bound is guaranteed.

PPO in Practice

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t\Big[\underbrace{L_t^{CLIP}(\theta)}_{\text{policy}} - c_1\underbrace{\big(V_\theta(s_t) - V_t^{\text{targ}}\big)^2}_{\text{value}} + c_2\underbrace{S[\pi_\theta](s_t)}_{\text{entropy}}\Big]$$

Each iteration:

1. N actors run $\pi_{\theta_{\text{old}}}$ for T steps each.
2. Truncated GAE: $\hat{A}_t = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l\delta_{t+l}$.
3. K epochs of minibatch Adam on L.
4. $\theta_{\text{old}} \leftarrow \theta$.

MuJoCo settings

horizon T	2048
epochs K	10
minibatch	64
$\gamma,\ \lambda$	0.99, 0.95
clip ϵ	0.2

K epochs on the same batch is the point: TRPO took **one** step per batch.

TRPO vs. PPO

	TRPO	PPO
Stay close by	KL constraint $\bar{D}_{\mathrm{KL}} \leq \delta$	clipped ratio $[1-\epsilon, 1+\epsilon]$
Solver	CG + line search (2nd order)	minibatch Adam (1st order)
Steps per batch	one	K epochs
Policy + value	separate networks	may share (one loss)
Advantages	Monte Carlo $\hat{Q}$	truncated GAE
States from	$s_0 \sim \rho_0$, then π_{old}	same — still no μ

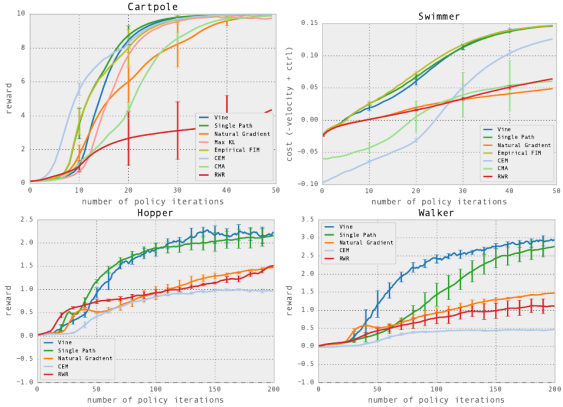
Kept from CPI and TRPO

- Surrogate on the **old** samples
- The ratio $\pi_\theta / \pi_{\theta_{\mathrm{old}}}$
- Small, controlled steps

Given up

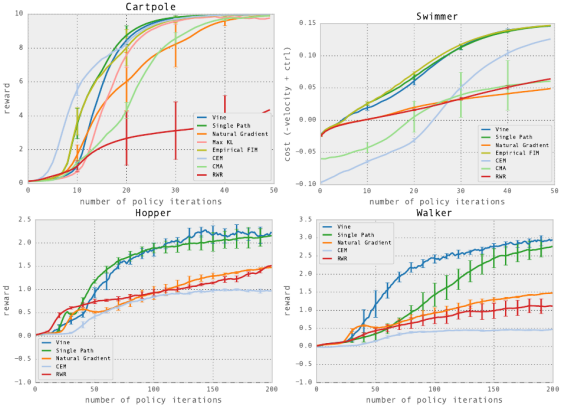
- Any improvement **bound**
- An exact limit on the KL
- (Monotone improvement: never had it in practice)

TRPO: Simulated Robots (MuJoCo)



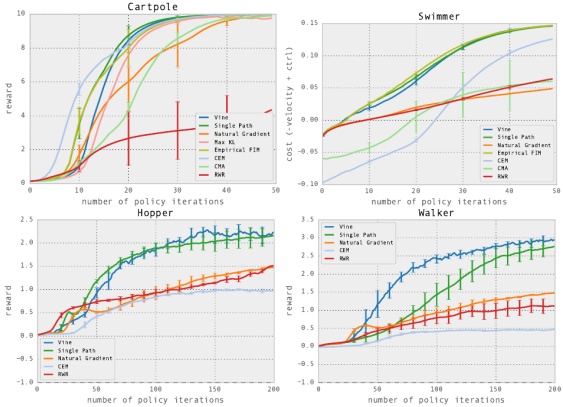
- Swimmer, hopper, walker: torques from joint states; neural-net policies.

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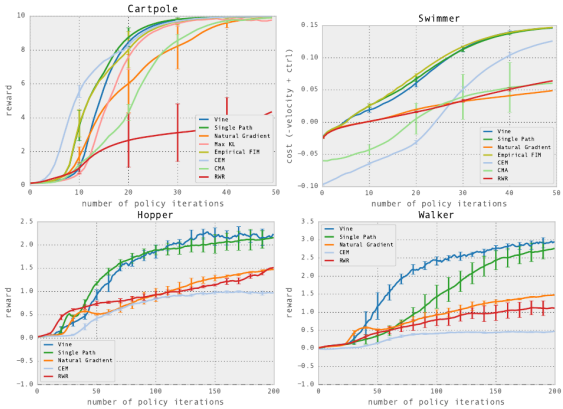
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- **Vine, single path**: solve all tasks, best scores.

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- **CEM, CMA** (derivative-free): poor on the larger networks.

TRPO: Simulated Robots (MuJoCo)



- Swimmer, hopper, walker: torques from joint states; neural-net policies.
- **Vine, single path**: solve all tasks, best scores.
- **Natural gradient** (fixed step): no hopping or walking gait.
- **CEM, CMA** (derivative-free): poor on the larger networks.
- **Max KL** (cart-pole only): a bit slower than mean KL — the simplification costs little.

Same $\delta = 0.01$ in every experiment.

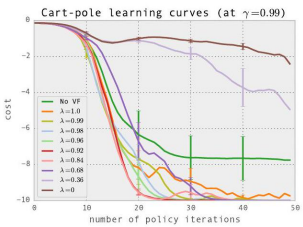
TRPO: Atari From Raw Pixels

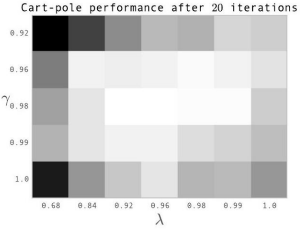
	B. Rider	Breakout	Enduro	Pong	Q*bert	Seaquest	S. Invaders
Random	354	1.2	0	−20.4	157	110	179
Human	7456	31.0	368	−3.0	18900	28010	3690
Deep Q-learning	4092	168.0	470	20.0	1952	1705	581
UCC-I (MCTS)	5702	380	741	21	20025	2995	692
TRPO single path	1425.2	10.8	**534.6**	**20.9**	**1973.5**	**1908.6**	568.4
TRPO vine	859.5	34.2	430.8	**20.9**	**7732.5**	788.4	450.2

Bold: TRPO above Deep Q-learning. One run per game (no error bars).

Not built for games — yet the **same** algorithm and settings as for the robots give reasonable scores everywhere.

GAE: Choosing γ and λ

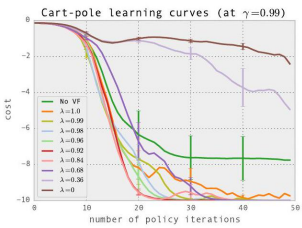


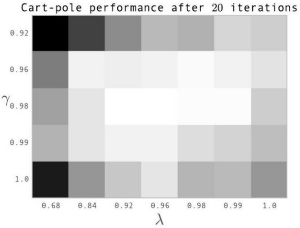


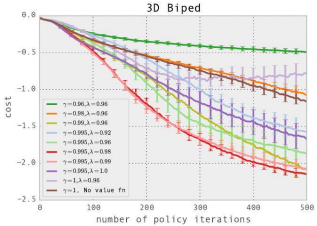
Cart-pole (top; cost, lower is better)

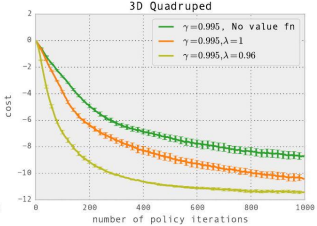
- best at intermediate $\lambda \in [0.92, 0.99]$
- $\lambda = 0$ and "No VF" much worse

GAE: Choosing γ and λ









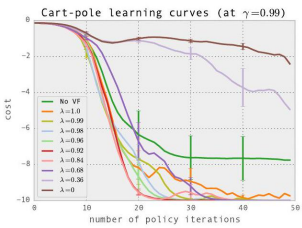
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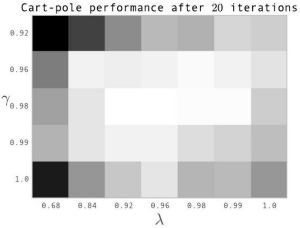
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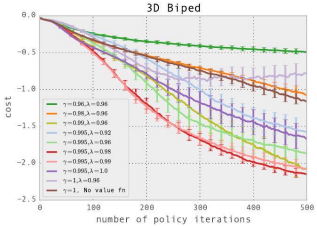
3D biped, quadruped (bottom)

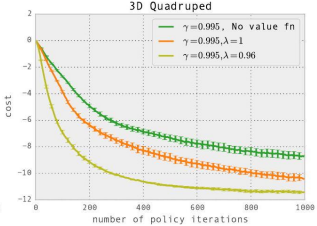
- best $\gamma \in [0.99, 0.995]$, $\lambda \in [0.96, 0.99]$
- biped runs after 1000 iterations $\approx$ 5.8 days of simulated time

GAE: Choosing γ and λ









Cart-pole (top; cost, lower is better)

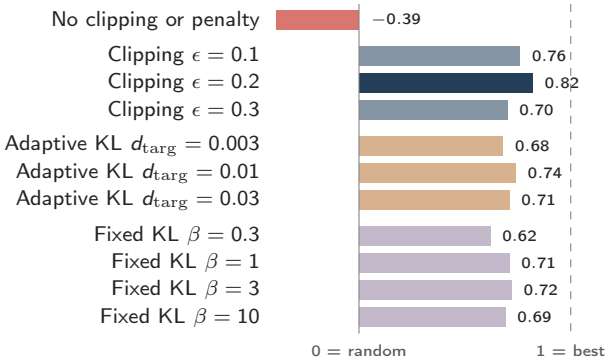
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3D biped, quadruped (bottom)

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best $\lambda <$ best γ: λ adds bias only if V is wrong.

Clipping vs. KL Penalty: the Ablation



Adaptive KL penalty

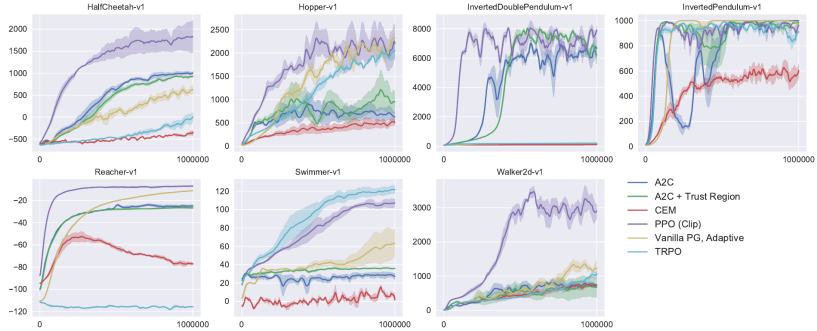
$$L^{KLPEN} = \hat{\mathbb{E}}_t\left[r_t \hat{A}_t - \beta\, D_{\mathrm{KL}}[\pi_{\theta_{\text{old}}}, \pi_\theta]\right]$$

After each update, with $d = \hat{\mathbb{E}}_t[D_{\mathrm{KL}}]$:

- $d < d_{\text{targ}}/1.5$: $\beta \leftarrow \beta/2$
- $d > 1.5\, d_{\text{targ}}$: $\beta \leftarrow 2\beta$

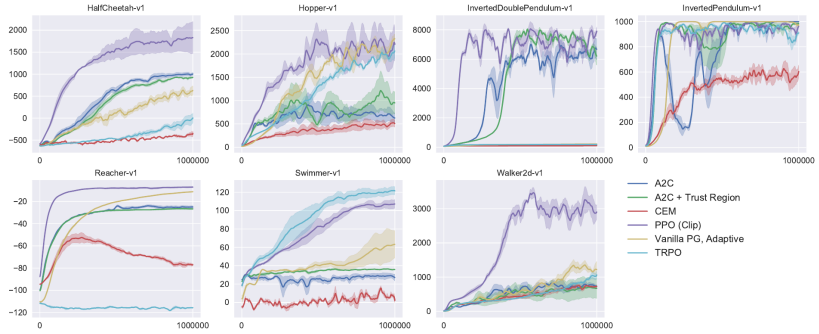
Clipping with $\epsilon = 0.2$ wins; the KL penalty stays a useful baseline.

PPO vs. Other Algorithms



MuJoCo, 1M steps: **PPO (clip)** best on almost all 7 tasks — above TRPO, A2C, CEM, vanilla PG.

PPO vs. Other Algorithms

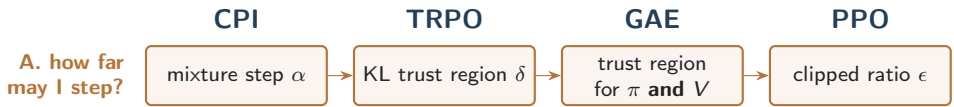


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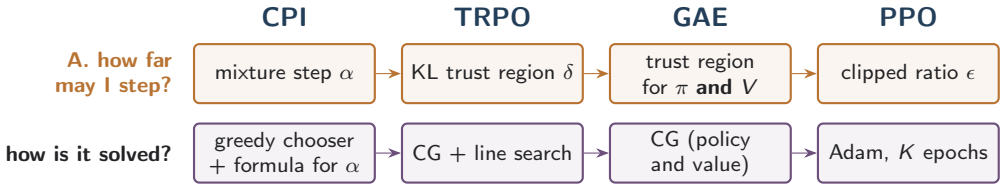
Atari, 49 games — games won:

	A2C	ACER	**PPO**
avg. over training	1	18	**30**
last 100 episodes	1	**28**	19

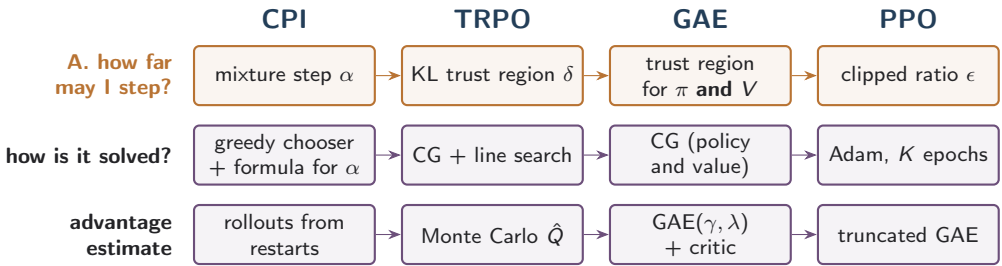
The Whole Story: CPI $\rightarrow$ TRPO $\rightarrow$ GAE $\rightarrow$ PPO



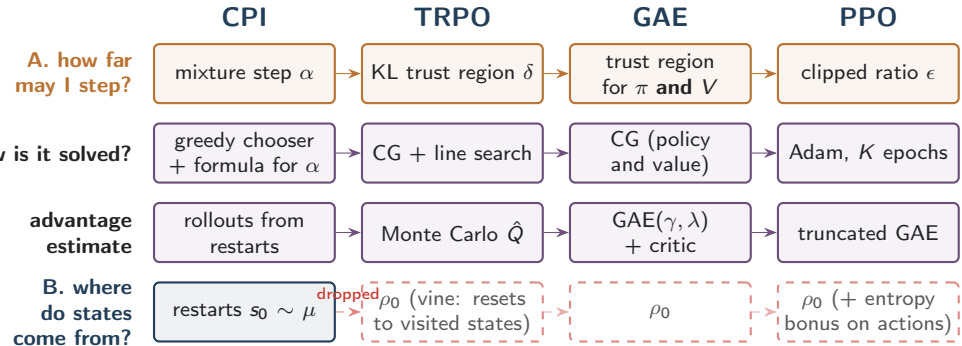
The Whole Story: CPI $\rightarrow$ TRPO $\rightarrow$ GAE $\rightarrow$ PPO



The Whole Story: CPI $\rightarrow$ TRPO $\rightarrow$ GAE $\rightarrow$ PPO



The Whole Story: CPI → TRPO → GAE → PPO



One Line per Paper

Paper	Problem it attacks	Main idea
CPI	greedy updates from approximate values can hurt	conservative mixture, step α from a bound
CPI	on-policy samples miss important states	exploratory restart distribution μ
TRPO	mixtures don't fit neural-net policies	step size $=$ a policy distance (KL)
TRPO (practice)	the theoretical penalty is far too conservative	mean-KL trust region, CG $+$ line search
GAE	$\hat{A}_t$ from returns is too noisy	critic V $+$ bias–variance knob λ
PPO	second-order step, one step per batch	clipped ratio $+$ minibatch SGD, K epochs

Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (1/3)

The conservative mixtures form a **one-parameter policy class** $\{\pi_\alpha\}_{\alpha \in [0,1]}$:

$$\pi_\alpha(a; s) = (1 - \alpha)\, \pi(a; s) + \alpha\, \pi'(a; s) \quad \Longrightarrow \quad \frac{\partial \pi_\alpha(a; s)}{\partial \alpha} = \pi'(a; s) - \pi(a; s)$$

So the policy gradient formula applies as is — α **plays the role of** θ:

Policy gradient, any $\theta \in \mathbb{R}^m$: $\nabla_\theta \, \eta_\mu = \dfrac{1}{1-\gamma} \sum_s d_{\pi_\theta, \mu}(s) \sum_a \nabla_\theta \pi_\theta(a; s)\, Q_{\pi_\theta}(s, a)$

$\theta := \alpha$

Mixture, at $\alpha = 0$ $(\pi_0 = \pi)$: $\left. \dfrac{\partial \eta_\mu}{\partial \alpha} \right|_{\alpha = 0} = \dfrac{1}{1-\gamma} \sum_s d_{\pi, \mu}(s) \sum_a \dfrac{\partial \pi_\alpha(a; s)}{\partial \alpha}\, Q_\pi(s, a)$

The factor $\frac{1}{1-\gamma}$ comes from our normalization: $d_{\pi,\mu}$ sums to 1 and Q_π carries a $(1 - \gamma)$.

Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (2/3)

Substitute the derivative of the mixture:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

Since $\sum_a \pi(a;s)\, Q_\pi(s,a) = V_\pi(s)$, $\sum_a \pi'(a;s) = 1$ and $A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$, the inner sum is:

$$\sum_a \pi'(a;s)\, Q_\pi(s,a) - V_\pi(s) = \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Back into the expectation — this is the policy advantage $\mathbb{A}_{\pi,\mu}(\pi')$:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] = \frac{1}{1-\gamma}\,\mathbb{A}_{\pi,\mu}(\pi')$$

Proof: α and θ Play the Same Role (3/3)

CPI: parameter $\alpha \in [0, 1]$

$$\eta_\mu(\pi_\alpha) = \eta_\mu(\pi) + \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

old-state surrogate exact to 1st order (this proof)

step size: α **itself**

generalize

TRPO: parameter $\theta \in \mathbb{R}^m$

$$\eta(\pi_\theta) = L_{\pi_{\theta_0}}(\pi_\theta) + o(\|\theta - \theta_0\|)$$

old-state surrogate exact to 1st order (TRPO Eq. 4)

step size (Thm. 1): $\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}})$

The proof never used that α is a mixing weight — only that it parameterizes the policy. TRPO keeps the first-order argument for any θ, and replaces "how far" (α) by a distance between policies.

Back to the policy advantage

Comparing With Bertsekas & Tsitsiklis (1/5): A Worst Case at Every State

$\bar{A}$ as in Lemma 6.1, $\Delta\eta = \eta_\mu(\pi') - \eta_\mu(\pi)$. Given: $\varepsilon = \max_s |\tilde{V}(s) - V_\pi(s)|$ and π' greedy w.r.t. $\tilde{V}$, i.e. $T_{\pi'}\tilde{V} \geq T_\pi \tilde{V}$,
where $(T_{\pi'} V)(s) = \mathbb{E}_{a\sim\pi'}\left[(1-\gamma)\mathcal{R}(s,a) + \gamma\, \mathbb{E}_{s'} V(s')\right]$, so $\bar{A} = T_{\pi'} V_\pi - V_\pi$.

$$T_{\pi'} V_\pi$$

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$$T_{\pi'}V_\pi \;\geq\; T_{\pi'}\tilde{V} - \gamma\varepsilon$$

V_π vs. $\tilde{V}$ at the next state: error $\leq \varepsilon$, times γ

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\begin{aligned}
T_{\pi'} V_\pi &\geq T_{\pi'} \tilde{V} - \gamma\varepsilon && V_\pi \text{ vs. } \tilde{V} \text{ at the next state: error} \leq \varepsilon, \text{ times } \gamma \\
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$$
\bar{A}(s) \;\geq\; -2\gamma\varepsilon \;\; \text{at every } s \quad\Longrightarrow\quad \Delta\eta = \frac{1}{1-\gamma}\,\mathbb{E}_{d_{\pi',\mu}}\big[\bar{A}\big] \;\geq\; -\frac{2\gamma\varepsilon}{1-\gamma}
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Comparing With Bertsekas & Tsitsiklis (1/5): A Worst Case at Every State

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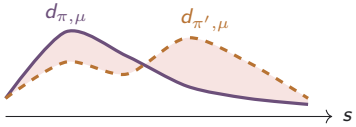
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$$

ε = **how wrong our values are**. It only limits how **negative** $\bar{A}$ can be; nothing positive is known, since π' is greedy for the *wrong* values. ($\varepsilon = 0$: $\bar{A} \geq 0$ everywhere — exact policy improvement.)

Comparing With Bertsekas & Tsitsiklis (2/5): Thm 4.1 at $\alpha = 1$

Split the new states into old states + shift:

$$(1-\gamma)\Delta\eta = \sum_s d_{\pi',\mu}(s)\,\bar{A}(s)$$

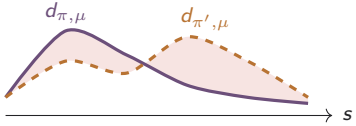


shaded: the shift — $\bar{A}$ unknown there.

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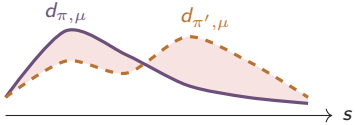


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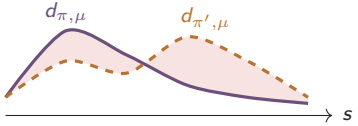
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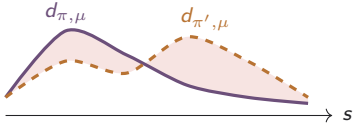
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- Same start $\mu \Rightarrow$ only $t \geq 1$ can differ: $\|d_{\pi',\mu} - d_{\pi,\mu}\|_1 \leq (1-\gamma)\sum_{t\geq 1}\gamma^t \cdot 2 = 2\gamma$

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$$\Delta\eta \;\geq\; \frac{\mathbb{A}}{1-\gamma} - \frac{2\gamma\varepsilon}{1-\gamma} \qquad = \text{Thm 4.1 at } \alpha = 1$$

Comparing With Bertsekas & Tsitsiklis (3/5): Why the Mixture Helps

Same split for $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi'$:

- Per state: $\bar{A}_{\text{new}}(s) = \alpha\,\bar{A}(s)$
  $\Rightarrow$ gain $\alpha\mathbb{A}$, $\max_s |\bar{A}_{\text{new}}| = \alpha\varepsilon$

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- Shift: π_{new} acts like π w.p. $1-\alpha$ each step,
  Pr(deviated by t) $\leq 1-(1-\alpha)^t$

$$\Rightarrow \quad \|d_{\pi_{\text{new}},\mu} - d_{\pi,\mu}\|_1 \leq \frac{2\alpha\gamma}{1-\gamma(1-\alpha)}$$

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- Multiply: $(1-\gamma)\Delta\eta \geq \alpha\mathbb{A} - \alpha\varepsilon \cdot \dfrac{2\alpha\gamma}{1-\gamma(1-\alpha)}$ $=$
  Thm 4.1

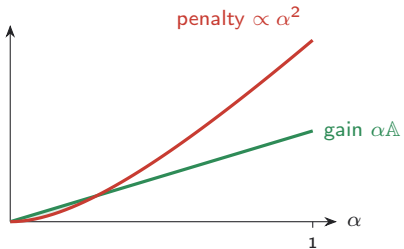
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Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$.

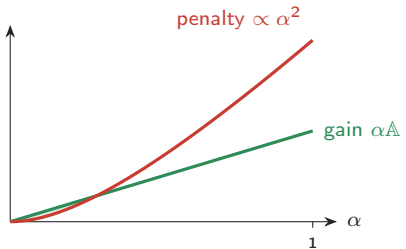
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- Multiply: $(1-\gamma)\Delta\eta \ge \alpha\mathbb{A} - \alpha\varepsilon \cdot \dfrac{2\alpha\gamma}{1-\gamma(1-\alpha)}$ = Thm 4.1



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$.

Small α: the gain wins. $\alpha = 1$: since $\mathbb{A} \le \varepsilon$, we need $\mathbb{A} > 2\gamma\varepsilon$ — impossible for $\gamma \ge \frac{1}{2}$. The positive term alone does not justify the full greedy step; **the small step does**.

Comparing With Bertsekas & Tsitsiklis (4/5): Same Penalty, Different ε

	B&T	K&L, Thm 4.1
$\varepsilon =$	$\max_s \lvert\tilde{V}(s) - V_\pi(s)\rvert$	$\max_s \lvert\mathbb{E}_{a\sim\pi'} A_\pi(s,a)\rvert$
Depends on	our estimate $\tilde{V}$	the pair π, π'
Measures	how wrong our values are	largest one-step gain/loss of π'
Step size	fixed: full greedy step	α (mixture weight)

Comparing With Bertsekas & Tsitsiklis (4/5): Same Penalty, Different ε

	B&T	K&L, Thm 4.1	TRPO, Thm 1
$\varepsilon =$	$\max_s \lvert\tilde{V}(s) - V_\pi(s)\rvert$	$\max_s \lvert\mathbb{E}_{a\sim\pi'} A_\pi(s,a)\rvert$	$\max_{s,a} \lvert A_\pi(s,a)\rvert$
Depends on	our estimate $\tilde{V}$	the pair π, π'	π only
Measures	how wrong our values are	largest one-step gain/loss of π'	scale of the advantages
Step size	fixed: full greedy step	α (mixture weight)	$\alpha = D_{\mathrm{TV}}^{\max}(\pi, \tilde{\pi})$

Comparing With Bertsekas & Tsitsiklis (4/5): Same Penalty, Different ε

	B&T	K&L, Thm 4.1	TRPO, Thm 1
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- **Common:** the price of not knowing $\bar{A}$ on the states the new policy visits.

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	B&T	K&L, Thm 4.1	TRPO, Thm 1
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Step size	fixed: full greedy step	α (mixture weight)	$\alpha = D_{\mathrm{TV}}^{\max}(\pi, \tilde{\pi})$

- **Common:** the price of not knowing $\bar{A}$ on the states the new policy visits.
- **Why it changed:** in B&T, π' is fixed by $\tilde{V}$, so only the estimate can go wrong. K&L ask a different question — *how far may we step?* — so ε measures the step, not the estimate.

Comparing With Bertsekas & Tsitsiklis (5/5): Where the Error Went, What TRPO Does

Where did B&T's approximation error go? Not into Thm 4.1's ε, but into

- the estimate of $\mathbb{A}$ from μ-restarts (sampling error, Hoeffding);
- the accuracy of the greedy chooser G_ε (CPI slide).

Both are **averages** under $d_{\pi,\mu}$ — much weaker than B&T's $\max_s$.

Back to Thm 4.1

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TRPO splits K&L's ε into a step and a scale:

$$|\bar{A}(s)| \;\le\; 2 \underbrace{D_{\mathrm{TV}}\big(\pi(\cdot|s), \tilde{\pi}(\cdot|s)\big)}_{\text{policy change} \le \alpha} \cdot \underbrace{\max_{s,a} |A_\pi(s,a)|}_{\text{advantage scale}}$$

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The step becomes a **distance between policies** that the algorithm controls ⇒ trust region.

◂ Back to Thm 4.1

From $\mathbb{E}_{s\sim d}$ to a Sum Over Time

$$\frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{\pi',\mu}}[\bar{A}(s)] = \frac{1}{1-\gamma}\sum_{s} d_{\pi',\mu}(s)\,\bar{A}(s)$$

Back to the lemma

From $\mathbb{E}_{s\sim d}$ to a Sum Over Time

$$\frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{\pi',\mu}}[\bar{A}(s)] = \frac{1}{1-\gamma}\sum_s d_{\pi',\mu}(s)\,\bar{A}(s)$$

$$= \sum_s \Big[\sum_t \gamma^t \Pr(s_t{=}s)\Big]\bar{A}(s)$$

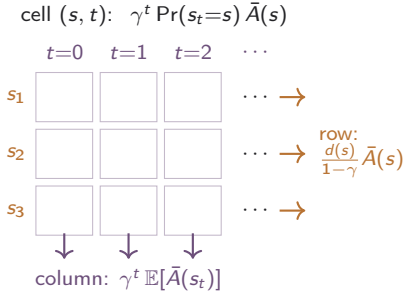
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$$= \sum_s \Big[\sum_t \gamma^t \Pr(s_t{=}s)\Big]\bar{A}(s)$$

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Sum the grid by rows (step 2)

or by columns (step 3): same total.

◂ Back to the lemma

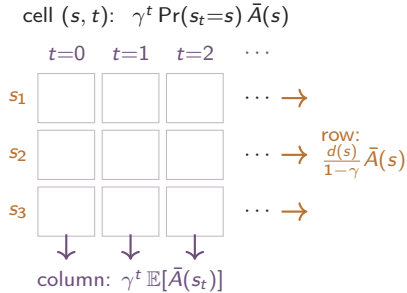
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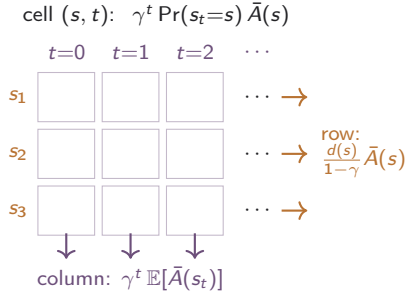
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$\mathbb{E}[\bar{A}(s_t)]$: over $s_t \sim p_t$, $p_t(s) = \Pr(s_t{=}s \mid s_0 \sim \mu, \pi')$ — the state at time t, **not discounted**.
$\mathbb{E}[\sum_t \cdots]$: over trajectories $s_0 \sim \mu$, $a_t \sim \pi'(\cdot|s_t)$, $s_{t+1} \sim P(\cdot|s_t, a_t)$.
The discount sits only in the weights γ^t.

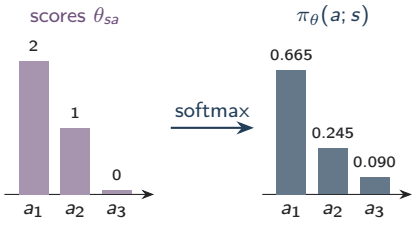


Sum the grid by rows (step 2)

or by columns (step 3): same total.

◂ Back to the lemma

The Gibbs (Softmax) Policy



One state s, three actions. Every action keeps **some** probability.

One parameter per state–action pair ("table lookup"):

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}$$

- **Why "Gibbs":** the Boltzmann distribution of physics, $p \propto e^{-E/T}$, with $\theta = -E/T$.
- **Always stochastic:** $\pi_\theta(a;s) > 0$; deterministic only as $\theta \to \pm\infty$.
- **Rich:** any policy with full support, in every state independently.
- **Simple score:** $\partial_{\theta_{sb}} \log \pi_\theta(a;s) = 1[a=b] - \pi_\theta(b;s)$

▸ Derivation

◂ Back to policy gradients

Deriving the Gibbs Score

$$\pi(a) = \frac{e^{\theta_a}}{\sum_c e^{\theta_c}}$$

Back to the Gibbs policy

Deriving the Gibbs Score

$$\pi(a) = \frac{e^{\theta_a}}{\sum_c e^{\theta_c}}$$

$$\log \pi(a) = \theta_a - \log \sum_c e^{\theta_c}$$

Back to the Gibbs policy

Deriving the Gibbs Score

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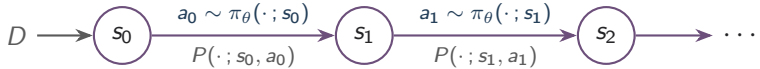
$$\log \pi(a) = \theta_a - \log \sum_c e^{\theta_c}$$

$$\partial_{\theta_b} \log \pi(a) = \underbrace{\partial_{\theta_b} \theta_a}_{1[a=b]} - \underbrace{\frac{e^{\theta_b}}{\sum_c e^{\theta_c}}}_{\pi(b)}$$

Back to the Gibbs policy

Why the Log-Derivative Trick Is Unbiased

θ sits in the **sampling distribution**, not in $R(\tau)$, so averaging $\nabla_\theta R$ gives nothing.



$$p_\theta(\tau) = D(s_0) \prod_t \pi_\theta(a_t; s_t)\, P(s_{t+1}; s_t, a_t)$$

$$\nabla_\theta \eta_D = \sum_\tau \underbrace{\nabla_\theta\, p_\theta(\tau)}_{p_\theta(\tau)\,\nabla_\theta \log p_\theta(\tau)} \qquad R(\tau) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\,\nabla_\theta \log p_\theta(\tau)\big]$$

$$\nabla_\theta \log p_\theta(\tau) = \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$$

(log turns $\prod$ into $\sum$; grey terms have no θ)

$\hat{g} = R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$, $\tau \sim \pi_\theta \;\Rightarrow\; \mathbb{E}[\hat{g}] = \nabla_\theta\, \eta_D$ **exactly**, and no need to know P.

Back to PG = SGA

CPI Through the SGD Lens

Take the table $\pi(a; s)$ itself as the parameter: CPI is a **first-order method** on η_μ.

	SGD (non-convex, smooth)	CPI
direction	$\hat{g} \approx \nabla J(w)$	greedy $\pi' = G_\varepsilon(\pi, \mu)$: best linearized direction
step	$w \leftarrow w + a_k\, \hat{g}$	$\pi \leftarrow (1-\alpha)\,\pi + \alpha\,\pi'$ (Frank–Wolfe step)
step size	$\sum_k a_k = \infty,\ \sum_k a_k^2 < \infty$	$\alpha \propto \hat{\mathbb{A}}$: slope over a smoothness bound
noise	averages out over many steps	each step accurate w.h.p. (Hoeffding)
stationary	$\|\nabla J(w_R)\| \le \epsilon$, random iterate	$\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$, last iterate
cost	$O(\sigma^2/\epsilon^4)$ samples	$72R^2/\varepsilon^2$ updates $\times$ $\tilde{O}(R^2/\varepsilon^2)$ rollouts
global?	$\times$ a stationary point can be bad	✓ Thm. 6.2, up to the mismatch factor

Back to the three questions

CPI Through the SGD Lens

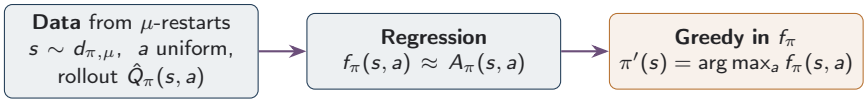
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Same skeleton: a bounded objective and a sure gain per step ⇒ few steps to a stationary point. CPI adds Thm. 6.2: **stationary** ⇒ **near-optimal**.

Back to the three questions

Building G_ε by Regression on Advantages



Required accuracy (Sec. 7.1):

$$L(f_\pi) = \underbrace{\mathbb{E}_{s\sim d_{\pi,\mu}}}_{\text{average over states}} \underbrace{\max_a}_{\text{worst action}} \left|A_\pi(s,a) - f_\pi(s,a)\right| \le \frac{\varepsilon}{2}$$

Why it suffices: with $e(s) = \max_a |A_\pi - f_\pi|$, $\hat{a} = \arg\max_a f_\pi$, $a^* = \arg\max_a A_\pi$:

$$A_\pi(s,\hat{a}) \ge f_\pi(s,\hat{a}) - e(s) \ge f_\pi(s,a^*) - e(s) \ge A_\pi(s,a^*) - 2e(s)$$

$$\mathbb{E}_{s\sim d_{\pi,\mu}}: \quad \mathbb{A}_{\pi,\mu}(\pi') \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - 2L \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - \varepsilon$$

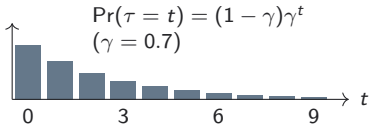
An **average** (ℓ_1) error over $d_{\pi,\mu}$, not the $\max_s$ (ℓ_∞) error of greedy DP (Eq. 3.1): rarely visited states may be fit badly.

Back to CPI

Stopping With Probability $1-\gamma$ Samples $d_{\pi,\mu}$

Run π from $s_0 \sim \mu$; after each step stop w.p. $1-\gamma$, independently of the trajectory. Return s_τ.

$\Pr(\tau = t) = \gamma^t(1-\gamma)$



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

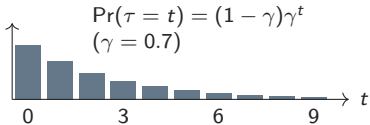
◂ Back to estimating 𝔸

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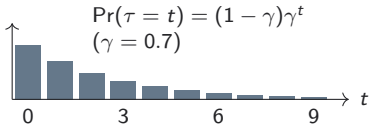
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◂ Back to estimating $\mathbb{A}$

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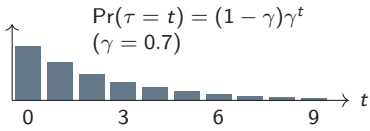
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Step 2 needs independence: τ does not look at the states.
The $(1-\gamma)$ in $d_{\pi,\mu}$ is what makes $\sum_t \Pr(\tau = t) = 1$.
γ = probability of surviving one more step;
$\mathbb{E}[\tau] = \gamma/(1-\gamma)$.



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

◂ Back to estimating 𝔸

Hoeffding's Inequality and the Estimate of $\mathbb{A}$

If $X_1, \ldots, X_k$ are **independent** and $X_i \in [a, b]$ a.s., then for $\bar{X} = \frac{1}{k} \sum_i X_i$:

$$\Pr\left(|\bar{X} - \mathbb{E}\bar{X}| > \Delta\right) \;\leq\; 2\exp\Big(-\frac{2k\Delta^2}{(b-a)^2}\Big)$$

◂ Back to estimating $\mathbb{A}$

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Conditions for x_i:

- **Independent:** each x_i comes from a fresh μ-restart.
- **Bounded:** $\hat{Q}_i \in [0, R]$, $|\pi' - \pi| \leq 1 \Rightarrow x_i \in [-n_a R,\, n_a R]$, $b - a = 2n_a R$.
- **Right mean:** $\mathbb{E}x_i = \mathbb{A}$ (uniform a, weight n_a); bias of a finite rollout kept below $\varepsilon/6$.

◂ Back to estimating $\mathbb{A}$

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$$\Pr\left(|\hat{\mathbb{A}} - \mathbb{A}| > \Delta\right) \;\le\; 2\exp\Big(-\frac{2k\Delta^2}{4n_a^2R^2}\Big) = 2e^{-k\Delta^2/(2n_a^2R^2)}$$

$$k \;\ge\; \frac{2n_a^2R^2}{\Delta^2}\log\frac{2}{\delta} \quad \text{gives error } \le \Delta \text{ w.p. } \ge 1-\delta$$

Back to estimating $\mathbb{A}$

Tool: Importance Sampling

The estimate of $\mathbb{A}$ needs a **sum** over actions, but each rollout tries **one** action.

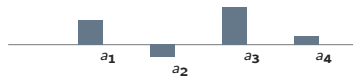
Sample from q, reweight by $w = p/q$

(needs $q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

$f(a)$

each a drawn w.p. $q = 1/n_a = \frac{1}{4}$



We can evaluate p pointwise, but only draw samples from q.

◂ Back to estimating $\mathbb{A}$

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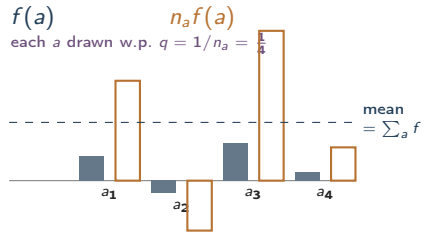
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A plain sum is the case $p \equiv 1$; CPI takes q uniform

$$\sum_a f(a) = \mathbb{E}_{a\sim q}\left[\frac{f(a)}{q(a)}\right] = \mathbb{E}_{a\sim \mathrm{unif}}\big[n_a f(a)\big]$$



Average of the orange bars under uniform a = sum of the blue bars.

◂ Back to estimating $\mathbb{A}$

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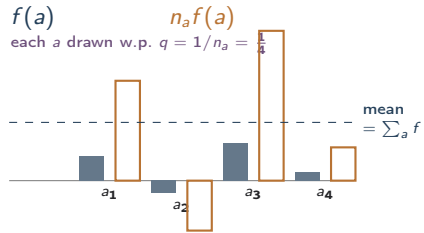
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Unbiased, but the weight stretches the range

$|f| \le R \;\Rightarrow\; |n_a f| \le n_a R \;\Rightarrow\; n_a^2$ in Hoeffding



One sampled bar can be n_a times taller ⇒ more samples needed. TRPO reuses this with $p = \pi_\theta$, $q = \pi_{\theta_{\text{old}}}$.

◂ Back to estimating $\mathbb{A}$

Why Uniform Actions? Choosing q

At a state $s \sim d_{\pi,\mu}$, the same target can be sampled two ways:

$$\sum_a (\pi' - \pi)(a; s)\, Q_\pi(s, a) \;=\; \underbrace{\mathbb{E}_{a\sim\text{unif}}\big[n_a(\pi' - \pi)(a; s)\, Q_\pi(s, a)\big]}_{\text{importance sampling, } p\equiv 1} \;=\; \underbrace{\mathbb{E}_{a\sim\pi'}\big[Q_\pi(s, a)\big] - V_\pi(s}_{\text{sample from } \pi'}$$

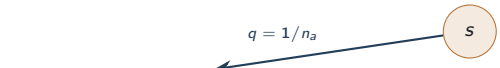


Back to estimating 𝔸

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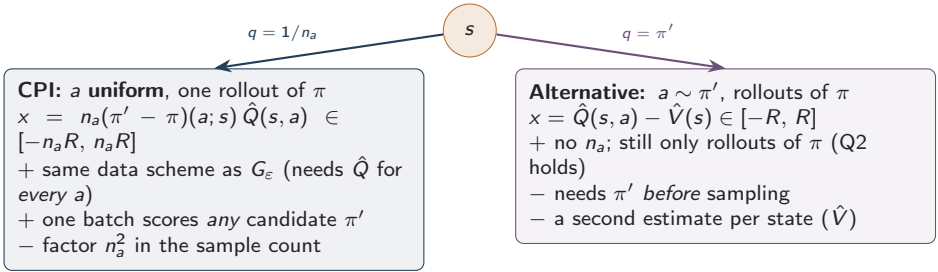
CPI: *a* **uniform**, one rollout of π
$x \;=\; n_a(\pi' - \pi)(a;s)\, \hat{Q}(s,a) \;\in$ $[-n_a R,\, n_a R]$
$+$ same data scheme as G_ε (needs $\hat{Q}$ for *every* a)
$+$ one batch scores *any* candidate π'
$-$ factor n_a^2 in the sample count

◂ Back to estimating 𝔸

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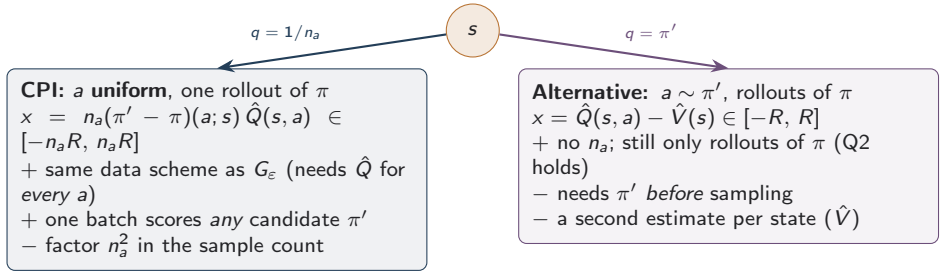


Back to estimating 𝔸

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Not plain IS with $q = \pi'$: weight $1/\pi'(a)$, but greedy π' is often deterministic, so $\pi'(a) = 0$ where $\pi(a)Q_\pi(s,a) \neq 0$ and the identity fails.

Back to estimating 𝔸

Continuous States and Actions?

Back to footnote 1: "replace the sums with integrals". **True** — no proof uses finiteness.

Why it carries over

- **Perf. difference**: telescoping, any space.
- **Coupling**: mass $1 - D_{\mathrm{TV}}$ on $\{a = \tilde{a}\}$ — also for densities.
- **Perturbation** (App. B): matrices $\to$ Markov operators, same norms.

What "integrals" hides

1. **Bounded** $\epsilon = \sup_{s,a} |A_\pi(s,a)| < \infty$.*
2. **max** $\to$ **sup** over **all** states, not just visited.
3. **Stochastic** policies: deterministic $\Rightarrow D_{\mathrm{KL}} = \infty$.
4. **Measurable** kernels; ρ_π is a measure.

* Often **vacuous** in continuous control: Gaussian actions with reward $-\|u\|^2$ (their Swimmer) give $\epsilon = \infty$. The practical algorithm drops the bound anyway (mean KL, fixed δ).

Back to Thm 1

One Problem, Several Known Algorithms

Objective	Constraint / penalty	Algorithm
Linear in θ	$\frac{1}{2}\|\theta - \theta_{\text{old}}\|^2 \leq \delta$	Vanilla policy gradient
Linear in θ	Quadratic KL, fixed step $\frac{1}{\lambda}$	Natural policy gradient
Full $L_{\pi_{\text{old}}}(\pi)$	none	Policy iteration
$L_{\theta_{\text{old}}}(\theta)$	$\bar{D}_{\text{KL}} \leq \delta$, enforced	**TRPO**

- NPG vs. TRPO: same direction $A^{-1}g$; TRPO **enforces** the KL constraint each step — matters on larger problems.
- Related: REPS constrains the joint $p(s, a)$; TRPO constrains only $\pi(a \mid s)$.

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